Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

Entwurf und Implementierung eines Echtzeit-Impedanzreglers für einen 7-achsigen kollaborativen Roboter

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Declaration of Authorship

I hereby declare that this master's thesis entitled

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Munich, 30.09.2026

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ORANGE The argument and presentation are sufficient for the present thesis. Minor polishing may still be possible.

YELLOW The content is relevant, but the explanation is dense, repetitive, weakly supported, or harder to follow than necessary.

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An unresolved concern can be raised beside the text as

The clearance gate holds the pose reached by the tool until operator confirmation. **comment:** "verify whether the same timing rule applies to the later gate"

A comment is a remark about the text, not a part of it. It appears only in this annotated copy and never in the submitted thesis.

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Abstract

ORANGE — sufficient. The abstract states the problem, implemented mechanism, principal measured ordering, null-space result, and scope limits without overloading the reader with the full parameter matrix.

Angular misalignment between a grinding tool and a surface can lead to edge-first contact and increased contact loads. In applications where the surface pose and tool mounting are subject to tolerances, the direction and magnitude of this misalignment are generally unknown before contact. This thesis investigates a Cartesian impedance control approach that uses the contact interaction itself to support the rotational alignment of the tool with a planar surface.

A real-time Cartesian impedance controller was implemented for a seven-degree-of-freedom (DOF) Franka Emika Panda. Translational and rotational compliance are defined relative to the surface geometry. The controller allows the virtual centre of compliance to be shifted away from the tool centre point (TCP). This shift introduces translation–rotation coupling into the Cartesian impedance, so that the commanded normal press can generate an additional moment that supports the required alignment rotation without prescribing an additional rotational trajectory.

The experimental investigation examined the influence of directional Cartesian stiffness and of the tangential position of the virtual compliance centre. The compliance-centre position produced the largest response variation over the tested parameter ranges. An assisting tangential displacement lies perpendicular to the required rotation direction. Reversing the sign of the initial angular deviation requires the displacement to be placed on the opposite side of the TCP, while changing the required tangent-plane rotation direction requires a corresponding change in the displacement direction.

Within the tested conditions, every non-zero tangential displacement that assisted

alignment was therefore specific to the initial misalignment direction and sign. The TCP-centred condition introduces no preferred tangential direction and is the direction-independent fixed centre of compliance among the tested positions for the investigated task. A displaced centre can instead be selected when the required alignment direction is known and additional rotational alignment authority is needed.

A complementary null-space study showed that projected damping reduced redundant joint motion. Singular-value conditioning was active while the commanded disturbance was applied and strongly suppressed the resulting net redundant displacement under the tested disturbance direction. This response therefore represents disturbance rejection during its application rather than a subsequent return of the redundant configuration.

Kurzfassung

ORANGE — sufficient. The German summary remains aligned with the English abstract in structure, certainty, findings, and limitations.

Eine Winkelabweichung zwischen einem Schleifwerkzeug und einer Fläche kann zu Kantenkontakt und erhöhten Kontaktbelastungen führen. Bei Anwendungen, in denen die Lage der Fläche und die Werkzeugaufnahme Toleranzen aufweisen, sind Richtung und Größe dieser Winkelabweichung vor dem Kontakt in der Regel nicht bekannt. Diese Arbeit untersucht einen kartesischen Impedanzregelungsansatz, bei dem die Kontaktinteraktion selbst die rotatorische Ausrichtung des Werkzeugs an einer ebenen Fläche unterstützt.

Für einen 7-achsigen Franka Emika Panda wurde ein kartesischer Echtzeit-Impedanzregler implementiert. Die translatorische und rotatorische Nachgiebigkeit wird relativ zur Flächengeometrie definiert. Das virtuelle Nachgiebigkeitszentrum kann gegenüber dem Werkzeugbezugspunkt (TCP) verschoben werden. Dadurch entsteht eine Kopplung zwischen Translation und Rotation innerhalb der kartesischen Impedanz, sodass die kommandierte normale Anpressbewegung ein zusätzliches Moment erzeugen kann, das die erforderliche Ausrichtungsdrehung unterstützt, ohne eine zusätzliche rotatorische Trajektorie vorzugeben.

Experimentell wurden der Einfluss der richtungsabhängigen kartesischen Steifigkeit sowie der tangentialen Lage des virtuellen Nachgiebigkeitszentrums untersucht. Innerhalb der geprüften Parameterbereiche führte die Lage des Nachgiebigkeitszentrums zur größten Variation der Ausrichtungsreaktion. Eine unterstützende tangentiale Verschiebung liegt senkrecht zur erforderlichen Drehrichtung. Wird das Vorzeichen der anfänglichen Winkelabweichung umgekehrt, muss die Verschiebung auf die gegenüberliegende Seite des TCP gelegt werden. Ändert sich die erforderliche Drehrichtung in der Tangentialebene, muss sich entsprechend auch die Richtung der Verschiebung ändern.

Unter den untersuchten Bedingungen war damit jede nicht verschwindende tangentiale Verschiebung, die die Ausrichtung unterstützte, von Richtung und Vorzeichen der anfänglichen Winkelabweichung abhängig. Die TCP-zentrierte Einstellung gibt keine bevorzugte tangentiale Richtung vor und stellt unter den getesteten Lagen das richtungsunabhängige feste Nachgiebigkeitszentrum für die untersuchte Aufgabe dar. Ein verschobenes Nachgiebigkeitszentrum kann dagegen gezielt eingesetzt werden, wenn die erforderliche Ausrichtungsrichtung bekannt ist und zusätzliche rotatorische Ausrichtungswirkung benötigt wird.

Eine ergänzende Nullraumuntersuchung zeigte, dass die projizierte Dämpfung redundante Gelenkbewegungen reduzierte. Die Singularwert-Konditionierung war während der kommandierten Störung aktiv und unterdrückte unter der untersuchten Störungsrichtung die daraus resultierende Netto-Verschiebung im Nullraum deutlich. Dieses Verhalten beschreibt somit eine Unterdrückung der Störwirkung während ihrer Einwirkung und keine nachträgliche Rückführung der redundanten Konfiguration.

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List of Abbreviations

ORANGE — sufficient. The abbreviations are separated from the symbol list and their printed forms are controlled centrally.

CoC	Centre of Compliance
CSV	Comma-Separated Values
DH	Denavit–Hartenberg
DOF	Degree of Freedom
EE	End-Effector
E-stop	Emergency Stop
FCI	Franka Control Interface
FCU	Franka Control Unit
F/T	Force/Torque
IP	Internet Protocol
LDLT	Lower–Diagonal–Lower-Transpose Factorisation
OS	Operating System
PREEMPT_RT	Real-Time Preemption Patch
ROS	Robot Operating System
SE(3)	Special Euclidean Group in three dimensions
SO(3)	Special Orthogonal Group in three dimensions
SVD	Singular Value Decomposition
TCP	Tool Centre Point

List of Symbols

ORANGE — sufficient. The notation is extensive but well organised, and the unit column makes the mixed translational and rotational quantities traceable.

Frames and Kinematics

$\{0\}$	Robot base frame	[-]
$\{\mathrm{EE}\}$	End-effector frame used by the robot model	[-]
$\{d\}$	Desired end-effector and TCP reference frame	[-]
$\{S\}$	Surface task frame, with axes t_1, t_2 and n_s	[-]
n	Number of joints ($n = 7$)	[-]
$q,\ \dot{q}$	Joint position / velocity vector	[rad], [rad/s]
p	Position / translation vector	[m]
R	Rotation matrix	[-]
T	Homogeneous transformation matrix	[-]
$J(q)$	Geometric Jacobian; linear rows above angular rows	[m/rad], [-]

Pose and Error

$p_{\mathrm{EE}},\ p_{\mathrm{TCP}}$	End-effector / tool-centre-point position; $p_{\mathrm{TCP}} = p_{\mathrm{EE}}$ in the implemented model	[m]
$R_{\mathrm{EE}},\ T_{\mathrm{EE}}$	End-effector orientation / transformation	[-]
$p_d,\ R_d,\ T_d$	Desired position / orientation / transformation	[m], [-]
ΔR	Current-to-desired relative rotation, $R_{\mathrm{EE}}^{\top} R_d$	[-]
ϕ	Orientation-error angle	[rad]
$u,\ [u]_{\times}$	Current-EE-frame orientation-error axis / its skew matrix	[-]
e_p	Translational (position) error	[m]
e_R	Base-frame rotational error used by the command	[rad]
$\dot{x}$	Cartesian velocity (twist)	[m/s], [rad/s]

Cartesian Impedance Gains

$K_p,\ D_p$	Generic translational stiffness / damping block; a frame or phase index is added when needed	[N/m], [N s/m]
$K_R,\ D_R$	Generic rotational stiffness / damping block; a frame or phase index is added when needed	[N m/rad], [N m s/rad]
$K_{p,t_1},\ K_{p,t_2},\ K_{p,n}$	Translational stiffness entries along t_1, t_2, and n_s	[N/m]
$K_{R,t_1},\ K_{R,t_2},\ K_{R,n}$	Rotational stiffness entries about t_1, t_2, and n_s	[N m/rad]

Dynamics and Torques

F	Cartesian wrench	[N], [N m]
$f,\ m$	Complete commanded Cartesian force / moment at the TCP, with $m = m_R + r_c \times f$	[N], [N m]
$f_n,\ f_t$	Normal / tangential component of the complete commanded force, $f_n = F_n n_s$ and $f_t = f - f_n$	[N], [N]
$K,\ D$	Full Cartesian stiffness / damping matrices	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
τ	Joint torque vector	[N m]
$\tau_c,\ \tau_g$	Coriolis / gravity compensation torque	[N m]
τ_{cart}	Cartesian impedance torque, $J^\top(q)F$	[N m]
τ_{cmd}	Complete commanded joint torque	[N m]

Surface and Contact Geometry

n_s	Calibrated surface-plane normal copied into the controller and used in the evaluation	[-]
$t_1,\ t_2$	Surface tangent directions (unit)	[-]
R_{surface}	Surface task frame $[\,t_1\ \ t_2\ \ n_s\,]$ in the base frame	[-]
$n_{\text{Tool,EE}},\ n_{\text{Tool,0}},\ n_d$	Tool-face normal expressed in the EE frame, fixed by calibration / the same direction expressed in the base frame, $n_{\text{Tool,0}} = R_{\text{EE}} n_{\text{Tool,EE}}$ / signed controller target direction	[-]
θ_{tilt}	Commanded orientation-offset rotation vector, $\theta_{\text{tilt}} = \theta_{t_1} t_1 + \theta_{t_2} t_2$	[rad]
p_s	Point on the configured surface	[m]
p_c	Centre of compliance	[m]
r_c	TCP-to-compliance-centre lever, $p_c - p_{\text{TCP}}$, as it enters the point-shift transformation and as the experimental plots and tables report it	[m]
$r_{c,t_1},\ r_{c,t_2},\ r_{c,n}$	Surface-frame components of r_c along t_1, t_2 and n_s	[m]
$r_{\text{Tool,EE}},\ p_{\text{Tool}}$	Selected tool-point offset from the TCP, expressed in the EE frame / selected tool-point position in the base frame	[m]
r_{Tool}	Selected tool-point offset from the TCP, $p_{\text{Tool}} - p_{\text{TCP}}$	[m]

m_R	Rotational-impedance contribution to the commanded moment, $K_R e_R - D_R \omega_{\mathrm{EE}}$, present at every lever setting	[N m]

Compliance and Alignment

$\mathrm{Ad}(r_c)$	TCP-to-compliance-centre point-shift adjoint	[-]
K_c, D_c	Block-diagonal stiffness / damping at the centre of compliance	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
F_c, F_{TCP}	Cartesian wrench at the centre of compliance / corresponding wrench at the TCP	[N], [N m]
$K_{\mathrm{TCP}}, D_{\mathrm{TCP}}$	Centre-of-compliance gains mapped by congruence to the TCP	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
θ_{t_1}, θ_{t_2}	Surface-frame components of the commanded tool orientation offset	[rad]
$\Delta\theta_i$	Measured signed end-effector rotation about surface tangent t_i from the beginning to the end of set-up	[rad]
$r_{c,t}$	Tangential part of the compliance-centre displacement, $r_{c,t_1} t_1 + r_{c,t_2} t_2$; its magnitude is written $\lVert r_{c,t} \rVert$	[m]
F_n, M_{t_i}	Signed commanded normal force / commanded TCP moment about surface tangent t_i	[N], [N m]

Null-Space and Singular Value Decomposition

J^+	Thresholded-SVD Moore–Penrose pseudoinverse	[rad/m], [-]
$N_q,\ N_\tau$	Joint-velocity / torque null-space projector	[-]
$\sigma_i,\ \sigma_{\min}$	Jacobian singular value / σ_6 conditioning indicator. The geometric Jacobian mixes linear and angular rows, so a singular value carries no single physical unit and its numerical value depends on the chosen Jacobian representation and length unit; it is used as a relative indicator under one fixed convention	[-]
v_7	Structural null direction of the seven-joint Jacobian	[-]
$\dot{q}_{\text{null}}$	Projected joint velocity in the numerical null space	[rad/s]
d_{null}	Null-space damping	[N m s/rad]
k_σ	Sigma-conditioning torque magnitude	[N m]
α_{probe}	Null-direction sampling angle	[rad]
$\Delta\sigma_{\text{min,dist}}$	Experimental change in $\sigma_{\min}$ over the disturbance interval, $\sigma_{\min}(9\,\text{s}) - \sigma_{\min}(5\,\text{s})$; distinct from the controller's Δ_σ	[-]
$\tau_d,\ \tau_\sigma,\ \tau_{\text{null}}$	Projected damping / sigma / total null-space torque	[N m]

E_N	Cumulative projected null-space motion; integrated projected joint-velocity magnitude during the disturbance interval, converted to degrees for reporting	[°]
v_{ref}	Fixed redundant comparison direction, taken from the condition without null-space torque	[-]
$\Delta q_{\text{null}},\ \Delta\eta_{\text{dist}}$	Net projected joint displacement over the disturbance interval / its signed component along v_{ref}	[rad]

Chapter 1

Introduction

1.1 Motivation

Industrial manipulators commonly use position control to track commanded trajectories. In surface-finishing applications, placement tolerances can cause the physical surface pose to differ from its programmed pose. The surface is positioned manually, and the tool is held in a gripper. Their residual angular misalignment is therefore unknown before contact. Rigid position control resists the corresponding corrective motion and can generate high contact loads.

Torque-controlled collaborative robots allow contact compliance to be specified directly and account for a growing share of industrial deployments [12, 24]. Joint torque sensing and a torque-level command interface determine how the robot yields under contact as well as where it moves. In surface finishing, this capability can reduce the loads caused by angular tool–surface misalignment. It can also permit the tool face to rotate towards the surface while maintaining the commanded normal press. The resulting alignment is evaluated relative to the physical surface.

Impedance control specifies this yielding behaviour as a dynamic relation between motion and interaction force [11]. Cartesian impedance control defines the relation for the end-effector pose. Its translational and rotational compliance can therefore be assigned independently along selected task directions. During edge-first contact, the wrench contains both force and moment components. Gains defined from the contact geometry allow part of this moment to rotate the tool towards the surface. The stiffness and damping matrices consequently determine both the contact response and the free-space tracking behaviour.

1.2 Related Work

The relevant literature comprises three areas: Cartesian impedance control, real-time implementation on torque-controlled robots, and the placement of the centre of compliance (CoC). The first area is Cartesian impedance control itself. The controller uses standard serial-manipulator kinematics [21, 3, 17]: homogeneous transformations for representing the end-effector pose and the Jacobian for relating joint motion to Cartesian motion. Khatib established the operational-space formulation, in which a Cartesian wrench F is mapped to joint torques τ through $J^\top$ [13]; this mapping is adopted in the controller. Hogan defined the impedance relation between end-effector motion and interaction force [11]. Albu-Schäffer et al. implemented Cartesian impedance control using joint-torque sensing [1], which corresponds to the torque-controlled interface used in this thesis.

The second area concerns the real-time implementation of Cartesian impedance control on torque-controlled robots. Mayr et al. implemented a modular Cartesian impedance controller for the Franka Emika Panda that combines real-time model access, null-space control, signal treatment, and command limiting [15]. The controller developed in this thesis uses the libfranka model interface to obtain the required robot state and model quantities during the real-time control loop. It also includes a selectable null-space torque τ_{null} for influencing the redundant joint motion without changing the commanded Cartesian task.

Recent work has applied compliant and force-controlled strategies directly to robotic surface finishing. Alt et al. combined perception, planning, and force-controlled execution for robotic surface treatment [2]. Min et al. integrated demonstrated motion and force skills with impedance control on a 7-DOF collaborative robot [16]. Wu et al. used adaptive variable impedance control to regulate the contact force during robotic grinding [27]. These studies address force-controlled surface treatment. The present work instead examines contact-induced angular alignment through directional Cartesian stiffness and a virtual CoC.

The third area concerns the location of the CoC. Fasse and Broenink formulated spatial impedance about a selected point, so that translation and rotation become coupled through the position of this point [5]. This formulation provides the theoretical basis for the compliance-centre shift used in this thesis. The same principle appears mechanically in remote-centre-compliance devices, in which elastic elements are arranged so that the tool behaves as if it rotates about a virtual point located away from the mechanism itself [18, 26]. More recent mechanically compliant and variable-compliance-centre strategies apply this principle to peg-in-hole insertion [25, 10]. In the mechanically compliant mechanisms, elastic bending produces the alignment motion rather than software control alone. In these approaches, the compliance is arranged so that contact forces reduce misalignment instead of increasing it. Most of these studies focus on insertion tasks and define the compliance centre through either the mechanical design or a strategy adapted to the hole geometry. In this thesis, an equivalent effect is created in software by shifting the virtual CoC away from the TCP. The independent effects of this position and the axis-specific stiffness during planar tool alignment are then investigated on a torque-controlled robot.

Suomalainen et al. experimentally investigated how compliance-centre placement and compliant-axis selection influence alignment in an assembly task [23]. Friedrich et al. subsequently proposed an active remote CoC that combines passive compliance with active force control [9]. These approaches further illustrate the role of compliance location in contact alignment. The present work examines a software-defined virtual compliance centre for planar tool–surface alignment.

1.3 Problem Statement

The angular relation between the tool and the surface plane is uncertain before contact because the surface placement and tool mounting both carry tolerances. The sign of the initial angle and its tangent-plane rotation direction are therefore unknown when a run starts. Under rigid position control, the contact geometry can generate high moments while the controller resists the corrective motion. In contrast, finite rotational compliance allows contact-induced rotation, whereas a displaced CoC adds a selectable coupling between translation and rotation.

The lever r_c is the displacement from the TCP to the virtual CoC. A tangential lever enters the commanded wrench through a cross product with the translational force and thereby selects an additional rotational moment. The direction and sign of this moment depend on the lever direction. A displaced centre chosen before contact therefore encodes a preferred alignment direction.

The investigation evaluates a fixed CoC for conditions in which the sign and tangent-plane direction of the initial angular misalignment are unknown. In this thesis, a direction-independent fixed centre denotes a centre position that can be selected without this prior information. The assessment is limited to the investigated task, parameter range, and tested centre positions. The term describes direction independence of the selected reference and carries no claim of global optimality.

Contact-induced alignment forms the set-up stage of the intended surface task. The tool approaches the surface with an angular offset, establishes contact, and rotates under the compliant interaction. Sustained tangential motion would follow after alignment. However, the reported measurements cover the contact set-up stage and end at the pre-grinding gate.

The alignment response depends on the directional stiffness, contact geometry, lever direction, robot configuration, and compliance of the tool mounting. The experiments therefore vary the selected controller parameters while keeping the calibrated geometry, the robot configuration, and the tool mounting unchanged. A commanded tool orientation offset represents the initial angular misalignment between the tool and the surface. Chapters 2 and 3 introduce the frame conventions and controller formulation used in the experiments.

1.4 Scope and Contributions

The scope covers contact set-up against one calibrated plane. Sustained grinding and adaptive selection of the compliance-centre position are outside the experimental evaluation. The main contributions are the following.

- A real-time Cartesian impedance controller with surface-relative directional gains and a shiftable virtual CoC was implemented on a Franka Emika Panda.
- The influence of directional stiffness and tangential compliance-centre position on contact-induced end-effector rotation was characterised experimentally.

- A projected null-space controller combining joint damping and singular-value conditioning was implemented and evaluated separately in Cartesian pose hold.

The contact study distinguishes a direction-independent fixed reference from a condition-dependent displaced centre. The former can be selected before the sign and tangent-plane direction of the initial misalignment are known. The latter adds rotational authority through the commanded moment $r_c \times f$ when the required direction is available.

1.5 Thesis Structure

Chapter 2 establishes the impedance law, frame conventions, and compliance-centre shift. Their real-time implementation on the Franka Emika Panda is described in Chapter 3. The experimental method and results follow in Chapters 4 and 5. Chapter 6 states the resulting conclusions and future work.

Chapter 2

Theoretical Background

This chapter presents the mathematical basis of the Cartesian impedance controller. The required frame conventions and pose errors are introduced first. The subsequent sections derive the kinematic mappings, the impedance wrench F, the compliance-centre transformation, and the null-space torque τ_{null}.

2.1 Reference Frames

The following right-handed coordinate frames are used in the theoretical formulation and the controller [21, 3].

- **Base frame $\{0\}$:** The base frame is fixed to the robot mounting surface. Its origin is the robot base reference point used by the kinematic model. For the upright Panda configuration used in this thesis, the z_0-axis is normal to the mounting surface, while the x_0- and y_0-axes lie in the mounting plane. Unless stated otherwise, Cartesian positions, directions, velocities, forces, and moments are expressed in this frame.
- **Joint frames $\{1\}, \{2\}, \ldots, \{n\}$:** These frames are attached to the successive links of the kinematic chain according to the Denavit–Hartenberg convention [4, 3]. Joint i rotates about z_{i-1}, and p_{i-1} denotes a point on this axis. The joint frames define the transformations between consecutive links and the indexing used in the forward kinematics and geometric Jacobian $J(q)$.
- **End-effector frame $\{\text{EE}\}$:** The end-effector frame is attached to the controlled TCP and moves with the tool. In the implemented robot model, the end-effector

origin and TCP are identified, so $p_{\mathrm{TCP}} = p_{\mathrm{EE}}$. The physical tool face is represented separately by its calibrated geometric offset from this frame. The position p_{EE} and orientation R_{EE}, both expressed relative to the base frame, define the current Cartesian pose of the robot. The axes x_{EE}, y_{EE}, and z_{EE} describe the current end-effector orientation. The configured tool approach axis is defined in this frame and is transformed to the base frame using R_{EE}.

- **Desired end-effector frame $\{d\}$:** The desired frame represents the commanded TCP pose. Its origin is p_d, and its orientation is R_d, both expressed relative to the base frame. The difference between the desired and current end-effector frames provides the translational and rotational errors used by the impedance controller.
- **Surface task frame $\{S\}$:** The surface task frame is tied to the configured surface rather than to the moving tool. Its origin is placed at the configured surface point p_s. Its orientation R_{surface} is defined by the two surface tangents t_1 and t_2 and the surface normal n_s:

$$R_{\mathrm{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \tag{2.1}$$

The directions t_1 and t_2 lie in the surface plane and are mutually orthogonal. The unit normal n_s is defined to point outward from the surface towards the free-space region above the plate. This sign convention is kept throughout the contact sequence, so motion towards the surface is along $-n_s$. The surface frame is used to describe the commanded tool orientation offset and to assign stiffness and damping values according to the two tangential directions and the normal direction. For the contact task, the task-frame orientation R_{task} equals the surface-frame orientation:

$$R_{\mathrm{task}} = R_{\mathrm{surface}}. \tag{2.2}$$

Figure 2.1 shows these frames together with the pose error between the current and the desired end-effector frame.

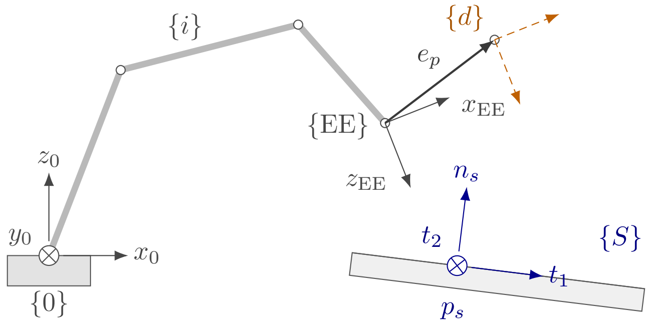


Figure 2.1: Frames used in the controller and the end-effector pose error.

2.2 Rigid-Body Transformations

The positions and orientations of the frames introduced in Section 2.1 are related through rotation matrices and homogeneous transformations.

The pose of frame $\{a\}$ in frame $\{b\}$ is represented by

$$^{b}T_{a} = \begin{bmatrix} ^{b}R_{a} & ^{b}p_{a} \\ 0^{\top} & 1 \end{bmatrix}, \qquad ^{b}v = {}^{b}R_{a}{}^{a}v. \tag{2.3}$$

The controller obtains the end-effector pose $T_{\text{EE}} = [R_{\text{EE}}, p_{\text{EE}}]$ directly from the libfranka robot state. Rotation matrices are used throughout; the pose is not converted to Euler angles.

2.2.1 Cartesian Pose Error

The desired end-effector transformation T_d is

$$T_d = \begin{bmatrix} R_d & p_d \\ 0^{\top} & 1 \end{bmatrix}, \tag{2.4}$$

where p_d and R_d are the desired TCP position and orientation, respectively. The current end-effector pose is denoted by T_{EE}.

The relative pose-error transformation T_{err} maps the current end-effector frame to the desired frame:

$$T_{\mathrm{err}} = T_{\mathrm{EE}}^{-1} T_d = \begin{bmatrix} R_{\mathrm{EE}}^{\top} R_d & R_{\mathrm{EE}}^{\top}(p_d - p_{\mathrm{EE}}) \\ 0^{\top} & 1 \end{bmatrix}. \tag{2.5}$$

When the current and desired poses coincide, $T_{\mathrm{err}} = I_4$. The rotational and translational blocks of Equation (2.5) therefore both represent zero error in this case.

Position error. The translational block of Equation (2.5) expresses the position error e_p in the current end-effector frame. The position error e_p used by the Cartesian impedance law is expressed in the base frame:

$$e_p = p_d - p_{\mathrm{EE}} \in \mathbb{R}^3. \tag{2.6}$$

A positive position error e_p therefore points from the current TCP position towards the desired TCP position.

Orientation error. The relative rotation ΔR is the rotational block of Equation (2.5):

$$\Delta R = R_{\mathrm{EE}}^{\top} R_d. \tag{2.7}$$

It represents the rotation required to carry the current end-effector orientation to the desired orientation. If both orientations coincide, then $\Delta R = I$.

For use in the impedance controller, ΔR is represented by a rotation angle ϕ and a unit rotation axis $u \in \mathbb{R}^3$ [21, 17]. The rotation-matrix and axis–angle representations can be converted in both directions. Given a unit axis u and an angle ϕ, the corresponding rotation matrix is obtained from Rodrigues' formula,

$$R(u, \phi) = I + \sin\phi\, [u]_{\times} + (1 - \cos\phi)[u]_{\times}^2. \tag{2.8}$$

For the relative rotation considered here, $R(u, \phi) = \Delta R$. The skew-symmetric matrix of the rotation axis is

$$[u]_\times = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix}, \qquad [u]_\times^\top = -[u]_\times. \tag{2.9}$$

Conversely, when the rotation matrix ΔR is known, the angle follows from

$$\mathrm{tr}(\Delta R) = 1 + 2\cos\phi, \tag{2.10}$$

and therefore

$$\phi = \arccos\left(\frac{\mathrm{tr}(\Delta R) - 1}{2}\right), \qquad \phi \in [0, \pi]. \tag{2.11}$$

Subtracting the transpose of ΔR isolates its skew-symmetric part:

$$\frac{1}{2}\left(\Delta R - \Delta R^\top\right) = \sin\phi\,[u]_\times. \tag{2.12}$$

For $\sin\phi \neq 0$, the rotation axis is obtained from

$$u = \frac{1}{2\sin\phi}\begin{bmatrix} \Delta R_{32} - \Delta R_{23} \\ \Delta R_{13} - \Delta R_{31} \\ \Delta R_{21} - \Delta R_{12} \end{bmatrix}. \tag{2.13}$$

Multiplying the unit axis by the angle gives the current-frame orientation error $e_{R,\mathrm{EE}}$. Rotating it into the base frame gives the controller orientation error e_R:

$$e_{R,\mathrm{EE}} = \phi u, \qquad e_R = R_{\mathrm{EE}} e_{R,\mathrm{EE}} \in \mathbb{R}^3. \tag{2.14}$$

The direction of e_R defines the axis of the required correction, while its magnitude is the angular error in radians. When the current and desired orientations coincide, $\phi = 0$ and $e_R = 0$. Together, $e_p = p_d - p_{\mathrm{EE}}$ and e_R are expressed in the base frame and point in the translational and rotational directions required to reduce the pose error.

The complete Cartesian error e stacks the position error e_p and orientation error e_R used by the impedance controller:

$$e = \begin{bmatrix} e_p \\ e_R \end{bmatrix} \in \mathbb{R}^6. \tag{2.15}$$

2.3 Kinematics of a 7-DOF Manipulator

Having defined the end-effector pose as the controlled variable, its relation to the joint configuration is introduced next.

For the joint configuration $q \in \mathbb{R}^7$, forward kinematics defines the end-effector pose $T_{\mathrm{EE}}(q)$. The Denavit–Hartenberg convention fixes the joint and frame indexing [4, 3], but the real-time controller obtains the pose and geometric Jacobian from the libfranka model interface.

The geometric Jacobian $J(q) \in \mathbb{R}^{6\times 7}$ maps joint velocity to the base-frame end-effector twist,

$$\begin{bmatrix} \dot{p}_{\mathrm{EE}} \\ \omega_{\mathrm{EE}} \end{bmatrix} = J(q)\dot{q}. \tag{2.16}$$

Its transpose maps the commanded Cartesian wrench to joint torque. These two mappings are the kinematic relations required by the impedance and null-space controllers.

2.4 Cartesian Impedance Control

The geometric Jacobian $J(q)$ is also used through its transpose to map a Cartesian wrench F to joint torques τ [13]. Before introducing this mapping, the desired Cartesian interaction behaviour is defined.

In this thesis, Cartesian impedance control is formulated as a spring–damper relation between the end-effector motion and the commanded wrench [11, 19, 15]. The Cartesian pose error e, defined in Section 2.2.1, produces a restoring wrench through the stiffness matrix K, while the Cartesian velocity $\dot{x}$ produces an opposing wrench through the damping matrix D. Their entries determine the translational and rotational compliance in the selected task directions [20]. Physically, the stiffness terms determine how strongly a pose error generates a restoring force or moment, while the damping terms oppose the corresponding Cartesian motion.

2.4.1 Impedance Wrench

The impedance wrench is evaluated in the robot base frame. The corresponding translational stiffness and damping matrices are $K_{p,0}, D_{p,0} \in \mathbb{R}^{3\times 3}$, while the rotational matrices are $K_{R,0}, D_{R,0} \in \mathbb{R}^{3\times 3}$. These matrices are not generally diagonal in the base frame because the direction-dependent gains are defined relative to the task frame. Their base-frame representation therefore changes with the task-frame orientation, even when the local gain values remain unchanged.

The commanded Cartesian force f is given by the translational spring–damper law [11, 15]

$$f = K_{p,0} e_p + D_{p,0} \left(\dot{p}_d - \dot{p}_{\mathrm{EE}}\right) \in \mathbb{R}^3, \tag{2.17}$$

where $\dot{p}_d$ and $\dot{p}_{\mathrm{EE}}$ are the desired and measured linear velocities expressed in the base frame. For a fixed position setpoint, $\dot{p}_d = \mathbf{0}$, and the damping term becomes $-D_{p,0}\dot{p}_{\mathrm{EE}}$.

The commanded Cartesian moment m is

$$m = K_{R,0} e_R - D_{R,0} \omega_{\mathrm{EE}} \in \mathbb{R}^3, \tag{2.18}$$

where ω_{EE} is the measured end-effector angular velocity expressed in the base frame. Since no angular reference velocity is commanded, the rotational damping acts directly against ω_{EE}.

The Cartesian wrench F stacks the force and moment:

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \begin{bmatrix} K_{p,0} e_p + D_{p,0} \left(\dot{p}_d - \dot{p}_{\mathrm{EE}}\right) \\ K_{R,0} e_R - D_{R,0} \omega_{\mathrm{EE}} \end{bmatrix} \in \mathbb{R}^6. \tag{2.19}$$

The wrench is ordered as force followed by moment, and all its components are expressed in the base frame. It is mapped to joint torques τ_{cart} through the Jacobian transpose in the following subsection.

Equation (2.19) defines the impedance about the TCP, with translation and rotation decoupled. Section 2.7 extends the same law to an impedance defined about a displaced centre of compliance.

2.4.2 Joint-Torque Control Law

By the principle of virtual work, the commanded Cartesian wrench F is mapped to joint torques τ_{cart} through the transpose of the geometric Jacobian $J(q)$ [13]:

$$\tau_{\text{cart}} = J^{\top}(q)F, \tag{2.20}$$

where $\tau_{\text{cart}} \in \mathbb{R}^n$ is the Cartesian impedance torque contribution and $J^{\top}(q) \in \mathbb{R}^{n\times 6}$. Substituting Equation (2.19) gives

$$\tau_{\text{cart}} = J^{\top}(q) \begin{bmatrix} K_{p,0}e_p + D_{p,0}\left(\dot{p}_d - \dot{p}_{\text{EE}}\right) \\ K_{R,0}e_R - D_{R,0}\omega_{\text{EE}} \end{bmatrix}. \tag{2.21}$$

The gain matrices in this expression are the base-frame matrices obtained from the selected task-frame gains. A change in the surface-frame orientation therefore changes the Cartesian wrench F and the resulting joint-torque contribution τ_{cart}, even when the local gain values remain unchanged.

2.4.3 Coriolis and Gravity Compensation in libfranka

In a general model-compensated torque controller, the commanded joint torque τ_{cmd} combines Cartesian impedance, gravity, and Coriolis/centrifugal compensation:

$$\tau_{\text{cmd}} = \tau_{\text{cart}} + \tau_g(q) + \tau_c(q, \dot{q}), \tag{2.22}$$

where $\tau_g(q)$ is the gravity torque and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal torque.

These terms are handled differently by the Franka Control Interface. The Coriolis vector provided by the libfranka model is added explicitly to the command because it is not compensated internally. A separate gravity term is not added. In external joint-torque mode, the robot compensates gravity and motor friction internally, and the commanded torque is interpreted as excluding gravity [7]. Adding $\tau_g(q)$ would therefore compensate gravity twice.

The implemented commanded torque $\tau_{\mathrm{cmd,impl}}$ is consequently

$$\tau_{\mathrm{cmd,impl}} = \tau_{\mathrm{cart}} + \tau_c(q, \dot{q}) = J^\top(q)F + \tau_c(q, \dot{q}), \tag{2.23}$$

which follows the official libfranka Cartesian impedance example [8]. The null-space torque τ_{null} introduced later is added to this expression to obtain the complete commanded torque in Equation (2.67).

2.5 Surface Task Frame

The contact experiment uses a task frame tied to the surface geometry. This frame is defined before the stiffness and damping matrices are introduced, so that the local gain entries can later be interpreted as normal and tangential values instead of as abstract base-frame directions.

Let the unit surface normal $n_s \in \mathbb{R}^3$, with $\|n_s\| = 1$, be expressed in the robot base frame. Its sign is chosen such that n_s points outward from the surface towards the free-space region above the plate. Consequently, the approach and set-up direction towards the surface is $-n_s$. The normal alone does not determine the frame. One rotational freedom about n_s remains, so the two tangent directions follow from a separate choice rather than from the surface geometry.

That choice is the surface-frame reference direction $a_s \in \mathbb{R}^3$, also expressed in the base frame. It is a configured direction and not a measured property of the surface. It need not lie in the surface plane, because only its component in the plane is used, and the single requirement is that it must not be parallel to n_s: a direction parallel to the normal leaves nothing to project. Two reference directions differing only by a multiple of n_s therefore define the same frame, whereas rotating a_s within the plane rotates t_1 and t_2 with it.

The first tangent vector is obtained by projecting a_s onto the plane orthogonal to n_s, and the second tangent vector is then obtained by a cross product. With $\tilde{t}_1$ denoting the unnormalised projected tangent, the construction is a Gram–Schmidt orthonormalisation step applied to the surface normal n_s and the surface-frame reference direction [21, 3]:

$$\begin{aligned}
\tilde{t}_1 &= a_s - n_s \left(n_s^\top a_s \right), \\
t_1 &= \frac{\tilde{t}_1}{\|\tilde{t}_1\|}, \\
t_2 &= n_s \times t_1.
\end{aligned} \tag{2.24}$$

Here, t_1 and t_2 are unit tangent vectors lying in the surface plane. This frame makes the gain directions physically meaningful: separate translational and rotational compliance can be assigned along the two surface tangents and along or about the surface normal instead of along arbitrary base-frame axes. Since the plane and its normal do not depend on a_s, a different reference direction changes which in-plane direction carries the name t_1 without changing the surface geometry. An axis-specific result therefore holds with respect to the configured reference direction, which for the reported experiments was the base-frame x-axis; the resulting tangents are given in Section 4.2.

The construction requires $\|\tilde{t}_1\| \neq 0$, and the behaviour of the implementation when the configured direction is close to the normal is described in Section 3.2.1. To match the implemented controller and its parameter order, the surface task frame collects the two tangents first and the normal last,

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \tag{2.25}$$

The column order is fixed: every surface-frame three-vector in the implementation uses $[\,\text{tangent } 1, \text{tangent } 2, \text{normal}\,]^\top$. In the gain representation below, the contact task uses this frame by setting $R_{\text{task}} = R_{\text{surface}}$. This makes stiffness values such as K_{p,t_1}, K_{p,t_2}, and $K_{p,n}$ local surface-frame quantities in that same order.

2.6 Stiffness and Damping Representation

Direction-dependent stiffness and damping gains are most conveniently defined in a task frame whose axes coincide with the desired principal compliance directions. The corresponding matrices are diagonal in that frame but are not generally diagonal when expressed in the robot base frame.

Let $R_{\text{task}} \in SO(3)$ denote the orientation of the task frame relative to the base frame. Its columns contain the task-frame axes expressed in the base frame. The local gains $K_{p,\text{task}}$, $D_{p,\text{task}}$, $K_{R,\text{task}}$, and $D_{R,\text{task}}$ are rotated into the corresponding base-frame gain matrices:

$$\begin{aligned} K_{p,0} &= R_{\text{task}} K_{p,\text{task}} R_{\text{task}}^{\top}, \quad D_{p,0} = R_{\text{task}} D_{p,\text{task}} R_{\text{task}}^{\top}, \\ K_{R,0} &= R_{\text{task}} K_{R,\text{task}} R_{\text{task}}^{\top}, \quad D_{R,0} = R_{\text{task}} D_{R,\text{task}} R_{\text{task}}^{\top}. \end{aligned} \tag{2.26}$$

For an impedance without translation–rotation coupling in the task frame, the full task-frame stiffness K_{task} and damping D_{task} are

$$K_{\text{task}} = \begin{bmatrix} K_{p,\text{task}} & 0 \\ 0 & K_{R,\text{task}} \end{bmatrix}, \qquad D_{\text{task}} = \begin{bmatrix} D_{p,\text{task}} & 0 \\ 0 & D_{R,\text{task}} \end{bmatrix}. \tag{2.27}$$

The block rotation matrix T_R applies the task-frame rotation to the translational and rotational blocks:

$$T_R = \operatorname{diag}\left(R_{\text{task}}, R_{\text{task}}\right), \tag{2.28}$$

The corresponding full base-frame stiffness K_0 and damping D_0 are

$$K_0 = T_R K_{\text{task}} T_R^{\top}, \qquad D_0 = T_R D_{\text{task}} T_R^{\top}. \tag{2.29}$$

The task-frame rotation changes the matrix representation without changing the physical directions assigned to the gains. For example, $K_{p,n}$ acts along the calibrated surface normal even when that normal is not aligned with a base-frame axis.

For the contact task, the task frame is the surface frame, whose orientation is

$$R_{\text{task}} = R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \tag{2.30}$$

For example, the local translational stiffness is defined as

$$K_{p,\text{task}} = \operatorname{diag}\left(K_{p,t_1}, K_{p,t_2}, K_{p,n}\right), \tag{2.31}$$

and its base-frame representation is

$$K_{p,0} = R_{\text{surface}} \operatorname{diag}\left(K_{p,t_1}, K_{p,t_2}, K_{p,n}\right) R_{\text{surface}}^{\top}. \tag{2.32}$$

The local rotational-stiffness entries K_{R,t_1}, K_{R,t_2}, and $K_{R,n}$ follow the same axis order:

$$K_{R,\text{task}} = \operatorname{diag}\left(K_{R,t_1}, K_{R,t_2}, K_{R,n}\right), \tag{2.33}$$

and the damping matrices are constructed analogously. A change in the configured surface orientation therefore changes the corresponding base-frame gain matrices even when the local gain values remain unchanged. During approach, contact set-up, and grinding, the reported contact runs used this surface-frame construction for both translational and rotational gain blocks. In the reported experiments, the phase-indexed numerical entries of Table 4.2 were assigned to $[t_1, t_2, n_s]$ before transformation to the base frame. The active gain matrices were based on the calibrated surface frame and were not rotated with the current end-effector orientation R_{EE}.

2.7 Centre of Compliance

The task-frame transformation of Equation (2.29) changes the directions along which the Cartesian stiffness and damping act, while leaving their reference point at the tool centre point (TCP). Independently of that orientation choice, the Cartesian impedance can be defined about a virtual CoC p_c. Shifting this point away from the TCP introduces translation–rotation coupling into the Cartesian impedance [17, 5, 22, 14]. The construction is applied during contact set-up, where it lets the normal pressing action generate a rotational moment as well as a translational contact load.

For a dual-arm assembly task, Suomalainen et al. experimentally showed that compliance-centre placement and compliant-axis selection affect the alignment response [23]. This task-level result motivates the placement study in this thesis. The point-shift adjoint below remains derived from the spatial-transformation formulation.

The displacement from the TCP to the CoC is

$$r_c = p_c - p_{\text{TCP}} \in \mathbb{R}^3, \qquad p_{\text{TCP}} = p_{\text{EE}}. \tag{2.34}$$

The vector r_c describes only the virtual shift of the impedance reference point.

Let $[r_c]_\times$ denote the skew-symmetric matrix satisfying $[r_c]_\times a = r_c \times a$. Under the local small-angle formulation, the translational error referred to the CoC is

$$e_{p,c} = e_p - [r_c]_\times e_R. \tag{2.35}$$

The negative sign follows from the selected lever convention. The vector from the TCP to p_c is r_c itself, and therefore

$$e_R \times r_c = -[r_c]_\times e_R.$$

The stacked error at the CoC collects $e_{p,c}$ and the rotational error e_R. The point-shift adjoint $\mathrm{Ad}(r_c)$ maps the corresponding TCP error onto it:

$$e_c = \begin{bmatrix} e_{p,c} \\ e_R \end{bmatrix} = \mathrm{Ad}(r_c) \begin{bmatrix} e_p \\ e_R \end{bmatrix}, \qquad \mathrm{Ad}(r_c) = \begin{bmatrix} I_3 & -[r_c]_\times \\ 0 & I_3 \end{bmatrix}. \tag{2.36}$$

The task-frame rotation and the point shift have different functions. The block rotation T_R of Equation (2.29) changes the principal directions of the impedance, whereas $\mathrm{Ad}(r_c)$ changes the point about which the impedance is defined.

After any required task-frame rotation, the translational and rotational stiffness and damping matrices are assembled at the CoC as

$$K_c = \begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}, \qquad D_c = \begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}. \tag{2.37}$$

All blocks in Equation (2.37) are expressed in the base-frame orientation. The compliance-centre wrench is $F_c = K_c e_c$, and transforming this power-conjugate wrench gives the corresponding TCP wrench $F_{\mathrm{TCP}} = \mathrm{Ad}(r_c)^\top F_c$. The matrices at the TCP therefore follow from the congruence transformations

$$K_{\mathrm{TCP}} = \mathrm{Ad}(r_c)^\top K_c \mathrm{Ad}(r_c) = \begin{bmatrix} K_p & -K_p[r_c]_\times \\ -[r_c]_\times^\top K_p & K_R + [r_c]_\times^\top K_p [r_c]_\times \end{bmatrix} \tag{2.38}$$

and

$$D_{\text{TCP}} = \text{Ad}(r_c)^\top D_c \text{Ad}(r_c) = \begin{bmatrix} D_p & -D_p[r_c]_\times \\ -[r_c]_\times^\top D_p & D_R + [r_c]_\times^\top D_p[r_c]_\times \end{bmatrix}. \tag{2.39}$$

The off-diagonal blocks introduce translation–rotation coupling. A translational position or velocity error can therefore contribute to the commanded moment, while a rotational error or velocity can contribute to the commanded force. For $r_c = 0$ the point shift disappears and the block-diagonal impedance at the TCP is recovered.

2.7.1 Commanded Moment

The Cartesian wrench generated by the impedance controller is $F = [f^\top,\ m^\top]^\top$, where f and m denote the complete commanded force and moment at the TCP.

Without a CoC displacement, the commanded moment is given by the ordinary rotational impedance,

$$m_R = K_R e_R - D_R \omega_{\text{EE}}. \tag{2.40}$$

A non-zero CoC displacement adds the translation–rotation coupling contribution $r_c \times f$, generated by the complete commanded force. The complete commanded TCP moment is therefore

$$m = m_R + r_c \times f. \tag{2.41}$$

This relation is exact for the point-shifted stiffness and damping matrices above. The controller evaluates the complete shifted spring–damper wrench and does not separately command an elastic or a damping coupling moment.

For $r_c = 0$ the coupling contribution vanishes and $m = m_R$. The TCP-centred condition is therefore a zero-coupling condition rather than a zero-moment condition. The rotational compliance remains finite and the commanded rotational impedance remains active.

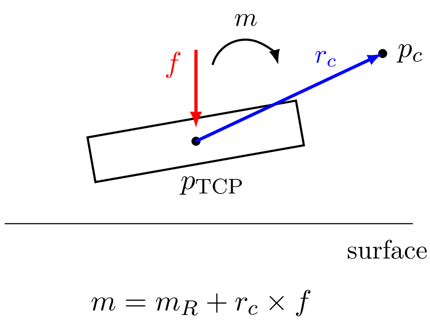


Figure 2.2: Centre-of-compliance displacement and commanded TCP moment. Shifting the virtual centre from the TCP by r_c introduces the coupling contribution $r_c \times f$.

2.7.2 Surface-Frame and Tangential Compliance-Centre Displacement

For the surface-contact task the CoC displacement is resolved in the surface frame $[t_1, t_2, n_s]$. Its surface-frame components are defined by

$$r_c = r_{c,t_1} t_1 + r_{c,t_2} t_2 + r_{c,n} n_s. \tag{2.42}$$

The tangential part collects the two in-plane components,

$$r_{c,t} = r_{c,t_1} t_1 + r_{c,t_2} t_2, \tag{2.43}$$

while $r_{c,n} n_s$ is the part along the surface normal. The quantities have to be kept apart: r_{c,t_1} and r_{c,t_2} are signed scalar coordinates, $r_{c,t}$ is the tangential vector they form, and its magnitude is written $\|r_{c,t}\|$. For a purely tangential displacement $r_{c,n} = 0$, and therefore $\|r_c\| = \|r_{c,t}\|$.

A non-zero displacement is specified by the frame in which its components are held constant as well as by its magnitude and direction. For a surface-fixed definition,

$$r_c = R_{\text{surface}} r_c^S, \qquad r_c^S = \begin{bmatrix} r_{c,t_1} \\ r_{c,t_2} \\ r_{c,n} \end{bmatrix}, \tag{2.44}$$

and r_c^S remains constant. For a tool-fixed definition the configured displacement remains constant in end-effector coordinates, and its base-frame representation rotates with the end effector. Section 3.4.3 describes both.

The commanded force enters through $r_c \times f$, and it decomposes exactly into a normal and a tangential component,

$$f = f_n + f_t, \qquad f_n = F_n n_s, \qquad f_t = f - f_n, \tag{2.45}$$

with $F_n = n_s^\top f$ the signed commanded normal force. The normal component is the projection of f onto n_s, and the tangential component is what remains of f in the surface plane. The complete CoC contribution is therefore

$$r_c \times f = r_c \times f_n + r_c \times f_t. \tag{2.46}$$

Both terms remain part of the complete commanded moment m.

The normal-press term simplifies. Substituting $r_c = r_{c,t} + r_{c,n} n_s$ and using $f_n \parallel n_s$, the normal component of the displacement drops out, $r_{c,n} n_s \times f_n = \mathbf{0}$, which leaves

$$r_c \times f_n = r_{c,t} \times f_n. \tag{2.47}$$

Only the tangential part of the displacement produces a moment from the commanded normal press. A displacement along the surface normal contributes nothing to this term. Consequently, the moment generated by the normal press depends on the tangential projection $r_{c,t}$, not on the total distance $\|r_c\|$ between the TCP and the CoC.

The right-handed surface frame satisfies

$$t_1 \times t_2 = n_s, \qquad t_2 \times n_s = t_1, \qquad t_1 \times n_s = -t_2. \tag{2.48}$$

Let the commanded orientation offset in the surface plane be

$$\theta_{\text{tilt}} = \theta_{t_1} t_1 + \theta_{t_2} t_2. \tag{2.49}$$

For a prescribed tangential magnitude $\|r_{c,t}\|$, the displacement used to assist the commanded alignment rotation is chosen perpendicular to the orientation-offset direction,

$$r_{c,t} = \|r_{c,t}\| \frac{\theta_{t_1} t_2 - \theta_{t_2} t_1}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}}. \tag{2.50}$$

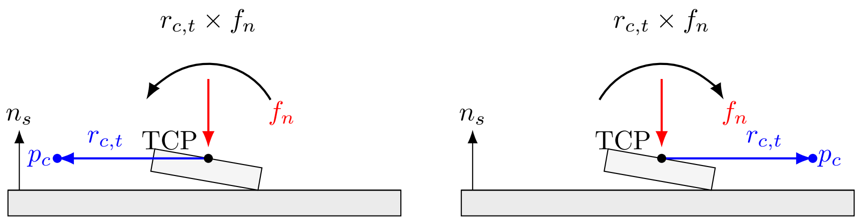
The choice fixes the direction; the magnitude remains a free parameter of the configuration. Using Equation (2.48), the normal-press contribution becomes

$$r_{c,t} \times f_n = \|r_{c,t}\| F_n \frac{\theta_{t_1} t_1 + \theta_{t_2} t_2}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}}. \tag{2.51}$$

During the set-up press $F_n < 0$, so this contribution acts opposite to θ_{tilt}. For the two principal directions the rule reduces to

$$r_{c,t} = +\|r_{c,t}\| t_2 \quad \text{for } +\theta_{t_1}, \qquad r_{c,t} = -\|r_{c,t}\| t_1 \quad \text{for } +\theta_{t_2}, \tag{2.52}$$

giving $r_{c,t} \times f_n = \|r_{c,t}\| F_n t_1$ and $r_{c,t} \times f_n = \|r_{c,t}\| F_n t_2$ respectively. Reversing the orientation offset reverses $r_{c,t}$ and therefore reverses this contribution. Figure 2.3 shows the geometry for the two signs.



(a) selected displacement (b) reversed displacement

$f_n = F_n n_s, \qquad m = m_R + r_c \times f$

Figure 2.3: Normal-force contribution to the compliance-centre coupling moment for opposite tangential displacements: (a) the displacement selected from the orientation offset; (b) the displacement with the opposite sign.

The complete commanded moment remains $m = m_R + r_c \times f$. The normal-force decomposition is used only to explain the directional effect of the CoC displacement. Tangential commanded-force components and damping effects remain contained in the complete wrench generated by the controller.

The coupling vanishes for $r_c = 0$. Conversely, if K_p is nonsingular, zero coupling implies $r_c = 0$. Since $\mathrm{Ad}(r_c)$ is invertible, the congruence transformation also preserves definiteness:

$$K_c \succeq 0 \implies K_{\mathrm{TCP}} \succeq 0, \qquad K_c \succ 0 \implies K_{\mathrm{TCP}} \succ 0. \tag{2.53}$$

The same statements apply to D_c and D_{TCP}. Negative off-diagonal entries therefore represent coupling terms rather than negative stiffness directions. Closed-loop robot–controller–contact behaviour with phase switching was assessed experimentally under the operating conditions reported in Chapter 4.

This formulation is a local small-motion model in which r_c is treated as fixed during one controller evaluation. A surface-fixed r_c^S therefore keeps its base-frame vector constant during set-up, whereas a tool-fixed displacement does not, and the controller updates the base-frame vector and the shifted gain matrices during each control cycle in that case.

The virtual CoC p_c is consequently a selected impedance reference point and is not interpreted as a measured physical pivot. It need not coincide with the physical contact location. The resulting contact response additionally depends on the finite rotational stiffness, the contact constraint, friction, the robot configuration, and tool-mount compliance.

2.8 Null-Space Torque

The Panda has one redundant joint direction when its 6×7 geometric Jacobian $J(q)$ has full row rank. The singular value decomposition (SVD) supplies two different quantities: v_7 spans the redundant direction, while $\sigma_{\min} = \sigma_6$ indicates the weakest task-producing mapping. A thresholded pseudoinverse gives the velocity and torque projectors. The controller then combines projected joint damping with a two-probe conditioning torque.

2.8.1 SVD and Null-Space Direction

The decomposition of the current Jacobian is

$$J(q) = U_J \Sigma_J V_J^\top, \tag{2.54}$$

where the six task-producing singular values satisfy $\sigma_1 \geq \cdots \geq \sigma_6 \geq 0$. At full row rank, the seventh right-singular vector satisfies

$$Jv_7 = 0, \tag{2.55}$$

and spans the one-dimensional kinematic null space. The conditioning indicator is

$$\sigma_{\min} = \sigma_6. \tag{2.56}$$

The vector v_7 and scalar σ_6 therefore have different roles. The first defines redundant motion; the second describes the weakest of the six task mappings. Because the geometric Jacobian mixes linear and angular rows, $\sigma_{\min}$ is used only as a relative indicator under one fixed Jacobian convention.

2.8.2 Pseudoinverse and Projectors

A relative tolerance ρ defines the retained singular values by $\sigma_i > \rho\sigma_1$. The corresponding Moore–Penrose pseudoinverse is

$$J^+ = \sum_{\sigma_i > \rho\sigma_1}^{6} \frac{1}{\sigma_i} v_i u_i^\top. \tag{2.57}$$

The joint-velocity projector extracts redundant motion,

$$N_q = I_7 - J^+ J, \qquad \dot{q}_{\text{null}} = N_q \dot{q}, \tag{2.58}$$

while the transpose projector applies a secondary joint torque without a first-order task component,

$$N_\tau = N_q^\top = I_7 - J^\top J^{+\top}. \tag{2.59}$$

With the thresholded SVD construction, the projector is orthogonal and symmetric. At full row rank it reduces to $N_q = N_\tau = v_7 v_7^\top$. The implemented separation is kinematic rather than inertia weighted; Cartesian task retention is therefore checked in the pose-hold experiment.

2.8.3 Projected Damping and Conditioning

Projected null-space damping opposes the redundant joint velocity,

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}. \tag{2.60}$$

The conditioning term samples two configurations along the current redundant direction. For the probe angle α_{probe}, these are

$$q_+ = q + \alpha_{\text{probe}} v_7, \qquad q_- = q - \alpha_{\text{probe}} v_7. \tag{2.61}$$

Their minimum singular values are

$$\sigma_+ = \sigma_{\min}(J(q_+)), \qquad \sigma_- = \sigma_{\min}(J(q_-)), \tag{2.62}$$

with the sampled difference

$$\Delta_\sigma = \sigma_+ - \sigma_-. \tag{2.63}$$

The direction selector applies a deadband δ_σ,

$$s_\sigma = \begin{cases} +1, & \Delta_\sigma > \delta_\sigma, \\ -1, & \Delta_\sigma < -\delta_\sigma, \\ 0, & |\Delta_\sigma| \leq \delta_\sigma, \end{cases} \tag{2.64}$$

and the conditioning torque is

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7). \tag{2.65}$$

Sampling q_+ and q_- selects the neighbouring configuration with the larger $\sigma_{\min}$. The finite probes are local tangent approximations to the nonlinear self-motion. The commanded magnitude k_σ is independent of the probe angle.

2.8.4 Complete Null-Space Torque

The complete secondary torque is

$$\tau_{\text{null}} = \tau_d + \tau_\sigma = -d_{\text{null}} N_\tau \dot{q} + k_\sigma N_\tau (s_\sigma v_7). \tag{2.66}$$

The damping term acts only while redundant motion is present. The conditioning term is an active configuration objective and can act at zero joint velocity. Its direction is invariant to the arbitrary sign returned for v_7, because reversing v_7 also exchanges the two probes and reverses s_σ.

2.9 Complete Joint-Torque Command

The complete joint-torque command τ_{cmd} combines the Cartesian impedance torque τ_{cart}, the null-space torque τ_{null}, and the Coriolis compensation τ_c:

$$\tau_{\text{cmd}} = \tau_{\text{cart}} + \tau_{\text{null}} + \tau_c(q, \dot{q}) = J^\top(q)F + \tau_{\text{null}} + \tau_c(q, \dot{q}). \tag{2.67}$$

Here, $\tau_{\text{cart}} = J^\top(q)F$ is the Cartesian impedance torque, τ_{null} is the secondary null-space torque, and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal compensation obtained from the robot model and added in the control callback. A separate gravity term is not added because gravity compensation is handled internally by the Franka Control Interface, as described in Section 2.4.3. The next chapter implements these torque contributions in the real-time control loop and combines them with the surface-relative geometry, phase-dependent references, and contact sequence.

Chapter 3

Controller Design and Real-Time Implementation

This chapter describes how the Cartesian impedance formulation of Chapter 2 is implemented as a real-time controller for the Franka Emika Panda. The controller is written in C++ using libfranka and Eigen and is executed through the Franka Control Interface (FCI) with a nominal control period of 1 ms.

The implementation is organised around the control flow required for the surface-alignment task. A surface-relative task representation first defines the surface frame, the tool orientation, and the point of the rectangular face used by the contact sequence. The real-time callback then updates the Cartesian reference, evaluates the impedance wrench, and maps it to joint torque. During set-up, the directional impedance can be shifted to a virtual CoC p_c, so that the commanded normal press also produces a moment. A secondary SVD-based null-space controller acts on the redundant joint motion without changing the Cartesian task to first order.

The architecture is introduced first, followed by the task geometry, real-time impedance calculation, phase-dependent gains and virtual CoC, contact sequence, null-space controller, and supporting real-time, safety, and recording functions. Numerical values used in the reported experiments are collected in Chapter 4 and Appendix C.

3.1 Controller Architecture

The software separates configuration and operator interaction from the real-time control calculation and from post-run data processing. The complete parameter set is read and validated before a controller session starts. The real-time callback then receives only the prepared configuration, preallocated state, robot-model interface, and non-blocking operator requests. File output and run evaluation are performed after torque control has stopped.

The main operating path is a phase sequence for tool orientation, surface approach, contact set-up, and grinding. Additional modes provide standard Cartesian pose hold, pose hold with the set-up impedance, and manual guidance. The same configuration system supplies the surface and tool geometry, phase-dependent Cartesian gains, damping settings, virtual CoC, null-space parameters, safety thresholds, and data-recording settings.

At program start, the robot connection is established and the robot model is loaded after error recovery and collision configuration. A separate keyboard thread receives operator requests. These requests are transferred through atomic variables, keeping terminal input outside the real-time callback. Before each new controller session, the configuration is read and validated again, which permits settings to be changed between runs without recompiling the program.

The torque path used during one impedance-controlled cycle is shown in Figure 3.1. The figure also indicates where the phase state, active Cartesian gains, robot model, and null-space contribution enter the command.

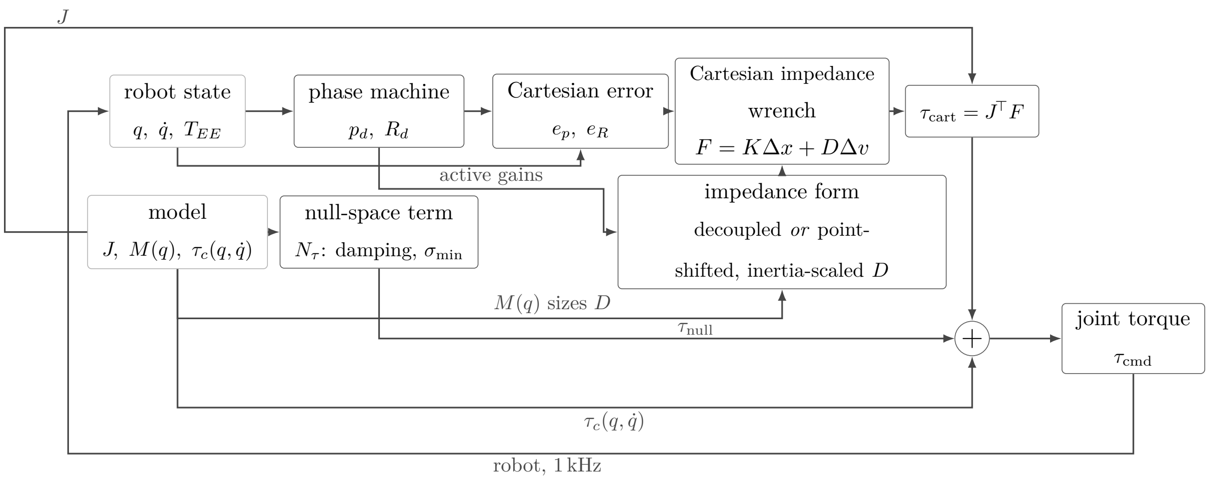


Figure 3.1: Torque contributions combined in one control cycle.

The following sections describe the task geometry, impedance calculation, phase logic, null-space controller, and real-time support functions in sequence.

3.2 Surface-Relative Task Representation

The contact sequence is formulated relative to the calibrated surface and to the known geometry of the tool face. The controller therefore constructs a surface task frame, a surface-relative tool orientation target, and a geometric reference point on the rectangular face before the set-up impedance is applied.

3.2.1 Surface Task Frame

The mathematical surface-frame construction is introduced in Section 2.5. In the implementation, the parameter set contains a point p_s on the surface, the surface orientation, and a surface-frame reference direction a_s used to define the first tangent. The resulting rotation has the fixed column order

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}.$$

The reference direction a_s is projected onto the plane orthogonal to n_s and normalised to obtain t_1. The second tangent is constructed as

$$t_2 = n_s \times t_1,$$

which completes the right-handed frame in robot-base coordinates. A fallback reference direction is used when the configured direction is approximately parallel to n_s.

This surface frame is used consistently for the approach direction, the surface-relative orientation command, directional Cartesian gains, the compliance-centre definition when a surface-fixed lever is selected, and the surface-resolved quantities used in the experimental evaluation. The descent direction is

$$-n_s,$$

from the free-space starting region towards the configured plane. The calibrated surface values used for the reported experiments are given in Chapter 4 and Appendix C.

3.2.2 Surface-Relative Tool Orientation

The tool normal is calibrated once in the end-effector frame. Its current direction in the robot base frame is calculated from the measured end-effector orientation:

$$n_{\text{Tool},0} = R_{\text{EE}} n_{\text{Tool,EE}}. \tag{3.1}$$

The calibrated surface normal n_s points outward from the surface towards the free-space region and therefore defines the flat tool-normal target. The commanded angular offsets about the two surface tangents are first combined into the rotation vector

$$\theta_{\text{tilt}} = \theta_{t_1} t_1 + \theta_{t_2} t_2 \in \mathbb{R}^3. \tag{3.2}$$

The vector θ_{tilt} specifies the commanded inclination relative to the surface. Its magnitude is the commanded tilt angle and its direction is the corresponding rotation axis, so the axis-angle relation of Equation (2.8) turns it directly into the rotation matrix R_{tilt}. Seating the tool face against the surface requires the tool normal to point into the plane, so the flat target is the inward surface normal,

$$n_{\text{flat}} = s_a n_s, \qquad s_a = -1, \tag{3.3}$$

with s_a the configured tool-axis target sign. The commanded rotation is applied to that axis, which gives the desired tool-normal direction,

$$n_d = R_{\text{tilt}} n_{\text{flat}} = R_{\text{tilt}} \left(s_a n_s \right). \tag{3.4}$$

For zero commanded tilt, $\theta_{t_1} = \theta_{t_2} = 0$, the rotation is the identity and $n_d = -n_s$.

The shortest rotation that aligns the captured tool direction with n_d is applied to the complete start orientation. The implementation can additionally prescribe a twist about the desired tool axis. The long axis of the rectangular face and the selected surface tangent are projected onto the plane normal to n_d, and the requested in-plane rotation is then applied about that axis. Because a centred rectangle is unchanged by a half-turn, the implementation selects the equivalent orientation closest to the captured start pose.

The construction separates the two orientation roles used later in the contact sequence: θ_{t_1} and θ_{t_2} define the commanded inclination relative to the surface, whereas the normal-axis twist defines the in-plane orientation of the rectangular face. The exact settings used for the measurements are stated in Chapter 4.

3.2.3 Tool Geometry and Geometric Clearance

The tool face is represented as a rectangle fixed in the end-effector frame. Its centre offset, half-width vector, and half-length vector define the four vertices

$$\{r_{\text{face,EE}} \pm r_{\text{width,EE}} \pm r_{\text{length,EE}}\}\,. \tag{3.5}$$

The tool geometry determines the point expected to reach the surface first. This point is denoted by p_{Tool}, the selected tool point, and is used as the contact reference during set-up. Depending on the tool orientation it corresponds to a leading corner, the midpoint of a leading edge, or the tool-face centre. It is distinct from the TCP and from the virtual CoC. Its position follows from the end-effector-frame offset $r_{\text{Tool,EE}}$,

$$p_{\text{Tool}} = p_{\text{TCP}} + R_{\text{EE}} r_{\text{Tool,EE}}, \tag{3.6}$$

where the tool centre point coincides with the reported end-effector origin in this implementation, $p_{\text{TCP}} = p_{\text{EE}}$. The offset $r_{\text{Tool,EE}}$ is fixed relative to the tool once the contact reference has been selected. Its base-frame representation $r_{\text{Tool}} = R_{\text{EE}} r_{\text{Tool,EE}} = p_{\text{Tool}} - p_{\text{TCP}}$ changes with the tool orientation. The compliance-centre shift $r_c = p_c - p_{\text{TCP}}$ of Section 2.7 is a different quantity: it is the virtual displacement used by the Cartesian impedance transformation, and it carries no tool geometry at all.

During tool orientation and surface approach, all four vertices are transformed to the base frame and compared along the descent direction. The vertex with the largest projection is geometrically closest to the surface in that direction. Vertices whose projected positions agree within the configured tie tolerance are grouped and averaged. A general orientation therefore selects a leading corner, two nearly equal neighbouring corners define a leading edge, and four nearly equal corners define the face centre.

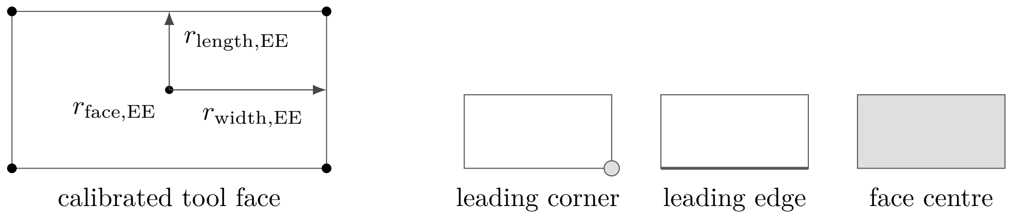


Figure 3.2: Tool face geometry and the selected leading feature.

The controller distinguishes the clearance of the foremost face point from the height of the averaged controlled point. The foremost-face clearance is used only to determine the transition from approach to set-up. The averaged point is the reference point used for the subsequent set-up motion.

At the geometric clearance transition, the implementation stores

- the selected tool-point offset in the end-effector frame;
- its projection onto the calibrated surface;
- the measured TCP pose and the orientation that will be held during set-up.

The selected face offset is then kept fixed during set-up. The desired contact-point position is generated relative to its projected surface point and converted back to the corresponding TCP reference. This allows the phase logic to command the geometrically relevant point on the tool face while the Cartesian impedance law remains formulated at the TCP.

3.3 Real-Time Cartesian Impedance Controller

One controller session prepares the task geometry and phase-dependent gains and then registers a joint-torque callback with libfranka. Before torque control begins, the current robot state is read once to initialise the desired pose, phase state, external-wrench reference, and preallocated recording buffer.

The callback receives the current robot state and elapsed cycle duration and returns the seven commanded joint torques τ_{cmd}.

Within each callback, the measured joint state, end-effector pose, end-effector Jacobian, robot-model quantities, and model-estimated external wrench are read first. The phase logic then updates the desired Cartesian reference. The controller evaluates the Cartesian error, selects the active impedance branch, maps the resulting wrench through $J^\top$, adds the secondary torque terms, records the required signals, and returns the final joint-torque vector.

3.3.1 State and Reference Update

The robot state provides the measured joint configuration q, joint velocity $\dot{q}$, and end-effector transformation. The libfranka model returns the 6×7 geometric Jacobian $J(q)$ of the configured end-effector frame together with the joint-space mass matrix and Coriolis terms. Multiplication of the Jacobian by $\dot{q}$ gives the measured Cartesian linear and angular velocities in the base frame.

The end-effector transformation supplied by the robot state is used directly for p_{EE} and R_{EE}; the real-time controller does not reconstruct this pose from a separate Denavit–Hartenberg chain.

The phase state machine supplies the desired TCP position p_d, desired linear velocity $\dot{p}_d$, and desired orientation R_d. Their meaning depends on the active phase: tool orientation changes the orientation reference while holding position, approach advances the position reference along $-n_s$, set-up advances the selected tool point into the surface, grinding maintains the normal press and can add a tangential sweep, and pose hold regulates a captured Cartesian pose. The detailed phase logic is described in Section 3.5.

3.3.2 Cartesian Error and Impedance Wrench

The translational error is the desired-minus-measured position error. The orientation error follows the convention introduced in Section 2.2.1. The current-to-desired relative rotation is converted to an axis–angle vector and expressed in the robot base frame:

$$\Delta R = R_{\mathrm{EE}}^\top R_d, \qquad e_R = R_{\mathrm{EE}} u \phi. \tag{3.7}$$

A rotational mask can be applied after this transformation by resolving e_R in the surface frame, selecting the enabled components, and transforming the result back to the base frame. The mask acts on the rotational spring error; the damping term continues to use the measured angular velocity.

The six-dimensional displacement and velocity errors are

$$\Delta x = \begin{bmatrix} e_p \\ e_R \end{bmatrix}, \qquad \Delta v = \begin{bmatrix} \dot{p}_d - \dot{p}_{\mathrm{EE}} \\ -\omega_{\mathrm{EE}} \end{bmatrix}. \tag{3.8}$$

For the decoupled impedance branch, the Cartesian wrench is evaluated from the translational and rotational blocks separately:

$$F = \begin{bmatrix} K_p e_p + D_p(\dot{p}_d - \dot{p}_{\mathrm{EE}}) \\ K_R e_R - D_R \omega_{\mathrm{EE}} \end{bmatrix}. \tag{3.9}$$

The same force–moment ordering is used for the coupled branch, but the 6×6 stiffness and damping matrices are first shifted from the selected CoC to the TCP as described in Section 3.4.3. The returned wrench is denoted

$$F = \begin{bmatrix} f \\ m \end{bmatrix} \in \mathbb{R}^6. \tag{3.10}$$

3.3.3 Joint-Torque Command

The Cartesian wrench is mapped to joint torque through

$$\tau_{\mathrm{cart}} = J^\top(q) F. \tag{3.11}$$

The SVD-based secondary controller supplies τ_{null}. During the dedicated Cartesian pose-hold disturbance study of Section 4.6, the callback also adds the internally commanded joint-torque-equivalent disturbance τ_{dist}. Outside this experiment, $\tau_{\mathrm{dist}} = 0$.

The Coriolis and centrifugal contribution $\tau_c(q, \dot{q})$ is obtained from the libfranka model. The complete impedance-controlled command is therefore

$$\tau_{\mathrm{cmd}} = \tau_{\mathrm{cart}} + \tau_{\mathrm{null}} + \tau_{\mathrm{dist}} + \tau_c(q, \dot{q}). \tag{3.12}$$

No separate gravity term is added because the Franka Control Interface handles gravity compensation for the external joint-torque command, as described in Chapter 2. The final torque vector is returned directly to libfranka; robot-side monitoring and the configured collision behaviour remain active.

3.4 Phase-Dependent Impedance and Centre of Compliance

The active Cartesian impedance is determined by three choices: the directional stiffness, the damping associated with those directions, and whether the impedance reference point is the TCP or a shifted virtual CoC. The controller constructs the directional gains first and applies the compliance-centre transformation afterwards.

3.4.1 Directional Cartesian Gains

Directional gains are defined in the frame that gives them a physical meaning for the active task. A diagonal gain G_{surface} expressed along $[t_1, t_2, n_s]$ is transformed to the base frame by

$$G_0 = R_{\text{surface}} G_{\text{surface}} R_{\text{surface}}^{\top}. \tag{3.13}$$

The base-frame matrix can then be applied directly to the base-frame Cartesian errors used by the control law. Surface-related phases use the calibrated surface frame, while Cartesian pose hold uses the base frame. The set-up translational frame remains configurable.

The set-up phase is the central case for the contact experiments. Its translational and rotational gains can be aligned with the surface frame so that normal and tangential stiffnesses can be selected independently. The specific matrices used in each experimental case are intentionally left to Chapter 4, where the varied and fixed entries are defined.

Set-up and grinding use the same source stiffness and damping blocks. The phase transition changes the generated Cartesian reference and the choice between the coupled and decoupled impedance branch. A phase gate can temporarily replace the active matrices with dedicated hold gains while the current reference is maintained.

3.4.2 Inertia-Scaled Damping

For phase groups configured to use inertia-scaled damping, the damping coefficients are calculated from the active directional stiffness and the Cartesian inertia associated with the current robot configuration. The joint-space inertia matrix

$$M(q) \in \mathbb{R}^{7\times 7} \tag{3.14}$$

is obtained from the libfranka model. The operational-space inertia in the robot base frame is evaluated from

$$\Lambda_0(q) = \Big(J(q)M(q)^{-1}J^\top(q) + \varepsilon I_6\Big)^{-1}, \qquad \varepsilon = 10^{-9}. \tag{3.15}$$

The inverse $M(q)^{-1}$ is not formed explicitly in the implementation. Instead, a lower–diagonal–lower-transpose (LDLT) factorisation of $M(q)$ is used to solve

$$M(q)Y(q) = J^\top(q), \qquad Y(q) = M(q)^{-1}J^\top(q) \in \mathbb{R}^{7\times 6}. \tag{3.16}$$

Here, $Y(q)$ is only the solution matrix of the linear system. The same operational-space inertia can therefore be evaluated numerically as

$$\Lambda_0(q) = (J(q)Y(q) + \varepsilon I_6)^{-1}. \tag{3.17}$$

The base-frame matrix is then transformed into the coordinate frame of the active gain block,

$$\Lambda_{\text{task}}(q) = T_R^\top \Lambda_0(q) T_R, \qquad \lambda_i(q) = [\Lambda_{\text{task}}(q)]_{ii}. \tag{3.18}$$

Thus, each directional stiffness entry k_i is paired with the corresponding Cartesian inertia $\lambda_i(q)$ of the same translational or rotational direction. The directional damping coefficient is then

$$d_i(q) = 2\zeta\sqrt{\lambda_i(q)k_i}, \tag{3.19}$$

where ζ is the configured directional damping factor. For a single decoupled direction with $\zeta = 1$, this expression corresponds to the critical-damping coefficient of that local second-order model.

The calculation is performed independently for the three translational and three rotational directions and is subject to the configured coefficient bounds and numerical validity checks. The resulting damping is cached by phase group and recalculated when that group is entered again, so the current robot configuration is reflected in $M(q)$, $J(q)$, and therefore in $\Lambda_{\text{task}}(q)$. If the calculation cannot be used, the configured damping matrix remains available as a fallback.

The reported surface-contact sequence used inertia-scaled damping in the phase groups stated in Chapter 4. Standard Cartesian pose hold is a separate case: its reported null-space trials used fixed hold damping rather than this inertia-scaled calculation.

3.4.3 Virtual Centre of Compliance

The theoretical point-shift formulation is derived in Section 2.7. In the implementation, the source translational and rotational gains are first expressed in the required task frame and assembled at a selected virtual CoC p_c. They are then shifted to the TCP.

The implemented lever convention is

$$r_c = p_c - p_{\text{TCP}}.$$

Two definitions are supported because the meaning of a fixed lever depends on the frame in which it is specified.

For a tool-fixed definition, the configured centre displacement is constant in end-effector coordinates. The corresponding base-frame lever is therefore updated from the measured end-effector orientation during every callback. It rotates in the base frame as the end effector rotates.

For a surface-fixed definition, the lever components are constant in $[t_1, t_2, n_s]$. Because the surface frame is fixed for one controller session, the resulting base-frame lever is constant during set-up. The configuration loader requires exactly one of the two definitions to be active when the virtual centre is enabled, and rejects a parameter set that selects both or neither.

The active base-frame lever is used to construct the point-shift adjoint. The source translational and rotational matrices are assembled into block-diagonal spatial gains at p_c, and the same congruence is applied independently to stiffness and damping:

$$K_{\mathrm{TCP}} = \mathrm{Ad}^{\top}(r_c) K_c \, \mathrm{Ad}(r_c), \qquad D_{\mathrm{TCP}} = \mathrm{Ad}^{\top}(r_c) D_c \, \mathrm{Ad}(r_c). \tag{3.20}$$

For a tool-fixed lever, r_c and therefore the shifted base-frame matrices are updated as the measured end-effector orientation changes. For a surface-fixed lever, the base-frame lever remains constant and the shifted matrices remain fixed as long as the source gains do not change.

The off-diagonal blocks generated by the point shift introduce translation–rotation coupling. A translational position or velocity error can therefore contribute to the commanded TCP moment, and a rotational error can contribute to the commanded TCP force. When $r_c = 0$, the point shift disappears and the spatial matrices reduce to the decoupled block-diagonal form.

This coupling is the mechanism used for contact-induced alignment during set-up. The desired orientation is held at the value captured at the geometric clearance transition, so no time-varying tipping trajectory is commanded. Instead, the selected tool point is advanced towards the surface. The resulting normal press acts through the finite Cartesian impedance; when a non-zero compliance-centre lever is enabled, the shifted impedance introduces a corresponding moment that can assist or oppose the contact-induced rotation. The measured response also depends on the directional stiffness, contact geometry, robot configuration, and physical tool–surface interaction.

The coupled branch is used only in the controller modes configured for a virtual CoC. Other phases use the decoupled Cartesian impedance. The exact cases that used the point-shifted branch in the reported experiments are specified in Chapter 4 and Appendix C.

3.5 Contact Sequence and Reference Generation

The main run sequence passes through tool orientation, surface approach, set-up, and grinding, in that order. The orientation phase can be omitted by configuration, and optional operator gates can hold the current reference between phases. Standard Cartesian pose hold, set-up-impedance hold, and manual guidance are separate operator modes. Figure 3.3 summarises the implemented phase flow.

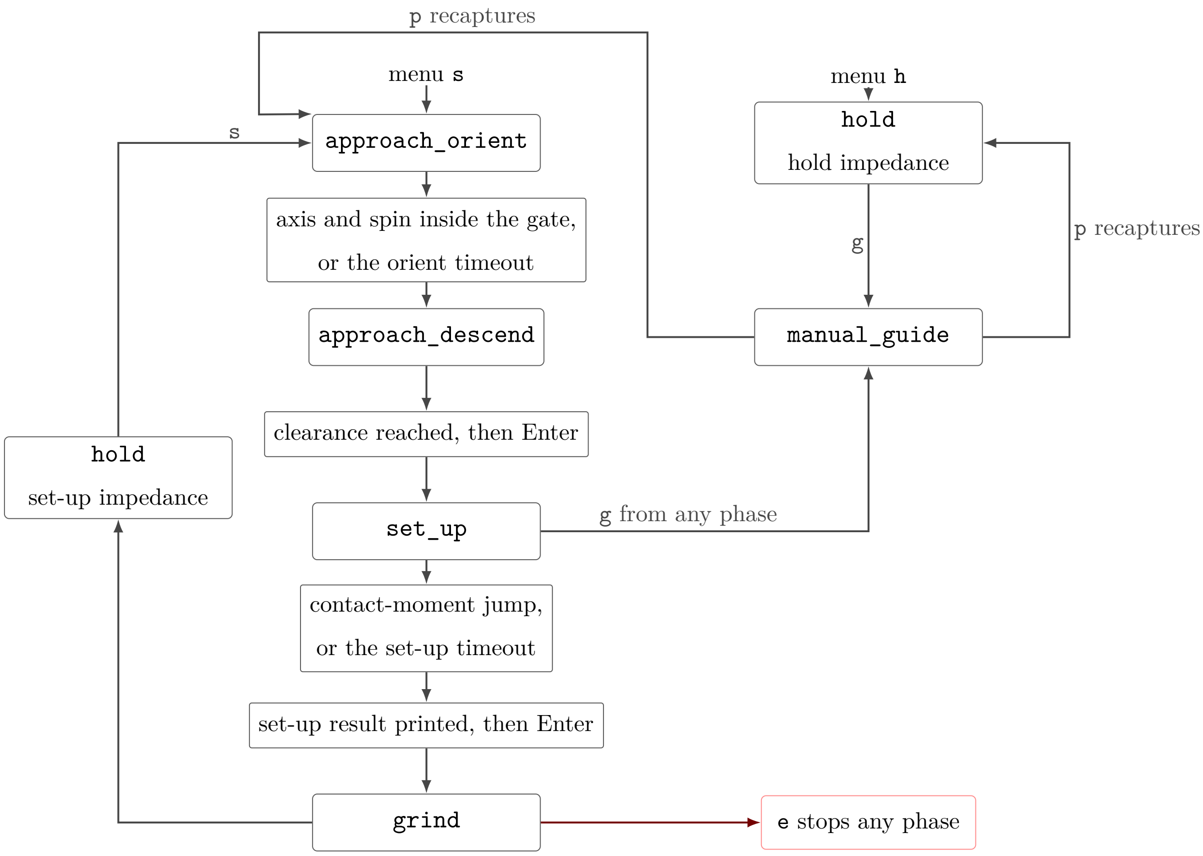


Figure 3.3: Run phase sequence and Cartesian pose hold.

3.5.1 Tool Orientation

The tool-orientation phase keeps the captured start position fixed while the desired orientation is advanced towards the surface-relative target of Section 3.2.2. The reference follows a rate-limited interpolation from the captured start orientation.

The transition is evaluated from the tool-axis error and, when the in-plane twist is enabled, the remaining spin error about the target tool axis. A configured minimum duration prevents immediate acceptance of a transient orientation, while a timeout provides an alternative transition path. The orientation reached at the end of this phase becomes the reference used during surface approach.

3.5.2 Surface Approach

During surface approach, the desired position advances from the captured start position along $-n_s$ with a rate limit and maximum travel. The rectangular tool geometry is evaluated continuously during this motion, so the controller always knows which corner or edge currently has the smallest geometric clearance to the surface.

The transition to set-up occurs when the foremost-face clearance reaches the configured geometric threshold. This is not a force-detected contact event.

At the clearance transition, the selected tool point is held constant, and the controller stores its projected surface point, the current TCP pose, and the orientation reference. If the optional pre-set-up gate is enabled, the controller first holds the captured pose until operator confirmation and then captures the set-up references from the current measured state. Reaching the maximum permitted approach distance before the geometric clearance is satisfied terminates the approach.

3.5.3 Set-Up

Set-up generates the contact press that is evaluated in the experiments. The calibrated normal n_s points outward from the surface, so the descent direction towards the surface is $-n_s$. At the clearance transition, the selected tool point is projected onto the calibrated plane. This projected point is denoted by $p_{\text{Tool},0}$ and provides the starting reference for set-up. The scalar coordinate $s_{\text{set}}(t) \geq 0$ describes the commanded displacement from this point towards the surface and is integrated at the configured speed and clamped at its configured endpoint. The desired position of the selected tool point is therefore

$$p_{\text{Tool},d}(t) = p_{\text{Tool},0} - s_{\text{set}}(t) n_s. \tag{3.21}$$

Here, $p_{\text{Tool},d}(t)$ is expressed in the robot base frame. The Cartesian impedance controller itself regulates the TCP, so this desired tool-point position must be converted into the corresponding desired TCP position.

The vector $r_{\mathrm{Tool,EE}}$ points from the TCP to the selected tool point and is expressed in the end-effector frame. At the clearance transition, the desired orientation for set-up is captured as R_{clr}. Expressing the stored tool-point offset in the robot base frame at this orientation gives

$$r_{\mathrm{Tool,clr}} = R_{\mathrm{clr}} r_{\mathrm{Tool,EE}}. \tag{3.22}$$

The desired tool-point position and desired TCP position are related by the same rigid-body geometry,

$$p_{\mathrm{Tool},d}(t) = p_d(t) + r_{\mathrm{Tool,clr}}. \tag{3.23}$$

Solving this relation for the desired TCP position gives

$$p_d(t) = p_{\mathrm{Tool},d}(t) - r_{\mathrm{Tool,clr}} = p_{\mathrm{Tool},d}(t) - R_{\mathrm{clr}} r_{\mathrm{Tool,EE}}. \tag{3.24}$$

Thus, $p_d(t)$ is the TCP reference required to place the selected tool point at $p_{\mathrm{Tool},d}(t)$ while maintaining the set-up orientation R_{clr}. The surface restricts the commanded motion of the tool point, so the resulting Cartesian position error generates the press.

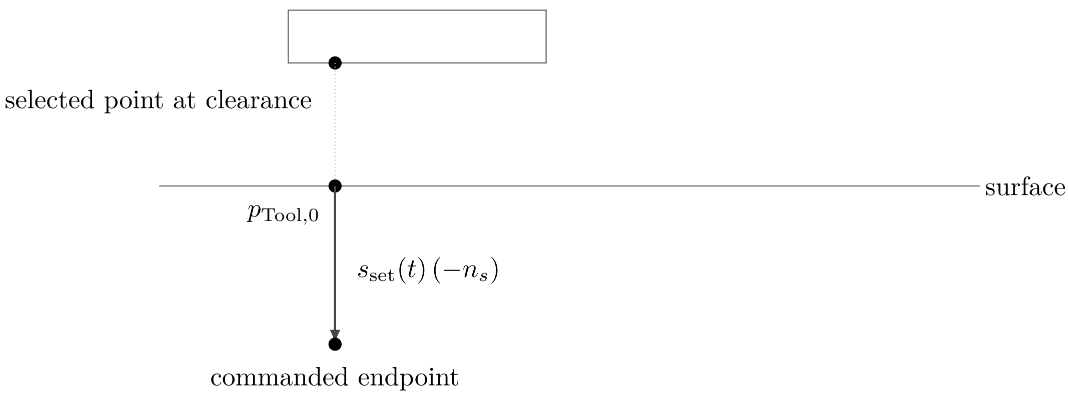


Figure 3.4: Set-up reference generation for the selected tool point.

The desired orientation remains equal to the orientation captured at the clearance transition. It therefore acts as a finite-stiffness rotational reference rather than as a rigid pose constraint. Contact-induced rotation can occur when the Cartesian wrench overcomes the corresponding rotational spring. When the virtual CoC is enabled, the point-shifted matrices of Section 3.4.3 additionally couple the translational press to the commanded moment.

The model-estimated external wrench supplied by libfranka is stored at the clearance transition for the optional wrench-change termination condition. After the configured minimum duration, set-up can terminate when the estimated external moment change exceeds the configured threshold or through the independent timeout. Every reported contact run terminated through the timeout, so this threshold did not affect the reported results.

If the pre-grinding gate is active, the controller remains in the set-up branch while waiting for operator confirmation. The coupled impedance therefore remains active when it was selected for that run. The exact transition settings and the termination behaviour observed in the reported experiments are documented in Chapter 4.

3.5.4 Grinding

The grinding phase maintains the normal set-up reference and can superimpose a smooth tangential motion along a selected surface tangent. Its impedance returns to the decoupled branch. No reported run entered this phase.

3.5.5 Pose Hold, Manual Guidance, and Run Termination

Standard Cartesian pose hold regulates a captured pose with the dedicated hold impedance. Set-up-impedance hold uses the set-up gain construction and can retain the virtual CoC. Manual guidance applies joint damping together with the model Coriolis contribution before the required references are captured again.

A controller session ends when the configured duration expires, an operator stop is requested, the maximum approach distance is exceeded, or libfranka terminates the motion because of a robot-side event. File writing and post-run evaluation occur only after the real-time callback has finished.

3.6 SVD-Based Null-Space Control

The Cartesian impedance defines the primary task. The Panda has one additional joint-space degree of freedom when the 6×7 task Jacobian has full row rank, so the implementation includes a secondary null-space controller for redundant motion. The theoretical projector, damping term, and two-point singular-value-conditioning strategy are derived in Section 2.8; this section describes their execution in the real-time callback.

For each impedance-controlled cycle, the full singular value decomposition of $J(q)$ is evaluated. The relative SVD tolerance is multiplied by the largest current singular value to obtain the threshold used in the Moore–Penrose pseudoinverse. Singular directions above the threshold are inverted; the remaining reciprocal entries are set to zero. The resulting joint-torque null-space projector N_τ is symmetrised numerically.

The projected joint velocity and damping torque are

$$\dot{q}_{\text{null}} = N_\tau \dot{q}, \qquad \tau_d = -d_{\text{null}} \dot{q}_{\text{null}}. \tag{3.25}$$

For the Moore–Penrose SVD construction used here the orthogonal projector is symmetric, so $N_\tau = N_q$ analytically. The implementation uses the numerically symmetrised N_τ both for the secondary torque and for the recorded projected joint velocity. The two symbols are kept apart in Section 2.8 only to distinguish the quantity the projector acts on.

The selectable secondary-torque modes are disabled, projected damping alone, singular-value conditioning alone, and their sum.

For singular-value conditioning, the current redundant right-singular direction v_7 is sampled in both signs. Two nearby joint configurations are formed at $q \pm \alpha_{\text{probe}} v_7$ and their Jacobians are evaluated using the current robot transformations. The smallest singular values at the two sampled configurations are compared. If their difference exceeds the configured deadband, the sign associated with the larger $\sigma_{\min}$ is selected; within the deadband, no conditioning torque is requested.

The selected direction is projected again through N_τ before the conditioning torque is returned. This keeps the commanded contribution in the numerical null space defined by the current pseudoinverse. The combined mode adds this contribution to the projected joint-velocity damping.

The surface-contact experiments used one fixed null-space configuration, while the dedicated pose-hold study varied the secondary terms to examine them separately. The corresponding numerical settings belong to the experimental method and are given in Chapter 4, Section 4.6, and Appendix C.

3.7 Real-Time Operation, Safety, and Data Recording

The functions around the control law are designed so that robot preparation, operator interaction, and file handling do not add blocking operations to the joint-torque callback. Safety monitoring remains active on the robot side, and the signals required for the experimental evaluation are written to preallocated memory during control.

3.7.1 Robot Preparation and Safety

Before a torque-controlled session begins, the software establishes the FCI connection, requests error recovery, applies the selected collision behaviour, and loads the robot model, in that order.

The robot can then be moved to a configured initial joint posture using a libfranka joint-motion generator, or the operator can start from a pose captured after manual guidance. The initial posture used in the reported experiments is documented in Appendix C rather than embedded in the controller description.

The complete joint-torque command of Equation (3.12) is returned directly to libfranka.

No additional application-side saturation or torque-rate limiter is applied after the Cartesian, null-space, disturbance, and Coriolis terms have been assembled. The configured robot-side collision/reflex monitoring therefore remains an independent robot-side protection and reflex mechanism. A robot communication, motion-generation, or torque-control error exits the callback through `franka::Exception`. Configuration, file, and numerical errors are handled separately through standard exceptions.

Gripper preparation and tool handling are completed before torque control and do not alter the Cartesian law or the experimental parameters.

3.7.2 Real-Time-Safe Data Recording

The controller stores the signals required for post-run analysis in a bounded in-memory ring buffer. The buffer memory is allocated before `robot.control` starts, so the callback does not grow the log dynamically. Each sampled row contains the measured state, active reference, calculated error and wrench, controller phase, secondary-controller state, and returned joint torque.

The complete column-level description is given in Appendix B. Values captured at the clearance transition refer to the geometric event of Section 3.5.2.

When the allocated capacity is reached, the write index wraps and new samples replace the oldest rows. This keeps the memory use bounded during real-time control. After the torque callback ends, the ring-buffer contents are written in chronological order to the CSV file. The experimental metrics calculated from them are defined in Chapter 4.

The controller structure described in this chapter is used throughout the reported experiments. Chapter 4 next specifies the physical system, calibration procedure, experimental parameter matrix, contact procedure, and evaluation quantities used to assess the implemented controller.

Chapter 4

Experimental Methodology

This chapter describes how the controller developed in Chapter 3 was evaluated experimentally. The methodology is organised in the order in which the experiments can be reproduced: the physical system is introduced first, followed by the surface and tool calibration, the common contact-test configuration and procedure, the main surface-contact matrix, the quantities used for evaluation, and the separate Cartesian pose-hold study for the null-space controller. Supporting checks of the compliance-centre geometry are reported in Appendix D.

The main data set consists of surface-contact measurements in which the commanded tool orientation offset, directional Cartesian stiffness, and virtual CoC p_c were varied under otherwise matched conditions. The primary controller-response quantity is the signed end-effector rotation during set-up. The TCP-height flatness classification is retained as an appendix check, while the commanded wrench supports the Case-D mechanism plot. A second, independent experiment evaluates the null-space terms under a repeatable internally commanded disturbance.

4.1 Experimental System

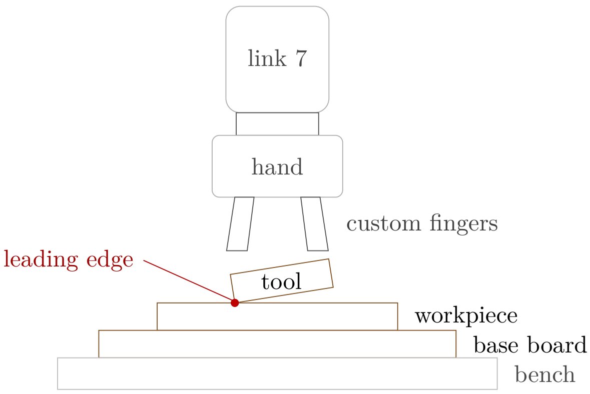
4.1.1 Robot, Tool, and Surface

The experiments were performed with a seven-degree-of-freedom Franka Emika Panda operated through the Franka Control Interface. Joint-torque commands were transmitted through libfranka at the nominal control rate of 1 kHz, corresponding to a sampling period of 1 ms. The real-time controller, Cartesian impedance law, contact geometry, and phase sequence are described in Chapter 3.

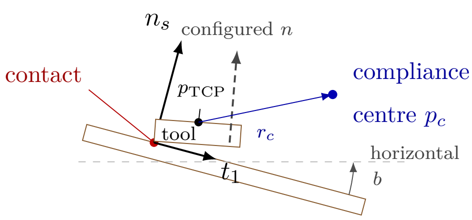
The rectangular grinding tool was held by the Franka Hand using custom gripper fingers. The tool face measured 40×120 mm, with the width along X_{EE} and the length along Y_{EE}. Its centre was located 20 mm along $+Z_{\mathrm{EE}}$ from the reported end-effector origin. The tool was clamped by custom pads screwed to the gripper fingers. Mechanical clearance in that clamping allowed the tool to rotate relative to the gripper by approximately $\pm 2°$ about y_{EE}, which corresponds approximately to the t_2 direction in the flat experimental configuration. The robot does not measure this relative rotation: it is absent from the joint and end-effector state. The $\pm 2°$ was measured in the unloaded condition, and the instantaneous relative tool–gripper rotation was not tracked during the contact experiments. It is therefore a mechanical property of the mounting rather than a calibrated measurement uncertainty.

Orientation quantities reconstructed from R_{EE} and the calibrated tool normal consequently describe the tool orientation under the assumption of a fixed tool-to-end-effector transformation, and any motion of the tool within the gripper is not included in them.

The surface was a plane plate mounted on a raised base board. All reported measurements used dry contact. The physical surface orientation and the tool normal were calibrated separately before the measurements, as described in Section 4.2.



(a) side elevation



(b) contact geometry

Figure 4.1: Experimental system and surface-frame contact geometry.

4.1.2 Real-Time Computer and Software

The controller computer ran Ubuntu 20.04.3 with the `5.9.1-rt20` Linux kernel and the `PREEMPT_RT` patch. The controller was compiled using `g++ 9.3`, C++14, and optimisation level `-O2`. Eigen 3.3.7 was used for matrix calculations and libfranka 0.7.1 for communication with the robot.

Before the measurements, the computer–robot connection was checked with the `communication_test` program supplied with libfranka 0.7.1 [6]. The program reports the fraction of control commands received by the robot and issues its lower communication-quality warning below an average command-success rate of 0.90. The experimental system passed this criterion. This check verifies communication quality for operation through the nominal 1 kHz interface; it is not a worst-case execution-time or scheduling-jitter measurement.

4.1.3 Operating and Safety Configuration

Before torque control started, the robot-side collision behaviour was configured with equal acceleration and nominal joint-torque thresholds of 140 N m. The six Cartesian force and moment channels used thresholds of 240 N or 240 N m, according to the component type. These thresholds supplied the Franka reflex mechanism and were kept unchanged throughout the reported measurements.

The complete run-specific controller configuration is given in Appendix C.

4.2 Surface and Tool Calibration

Two geometric quantities were required before the contact experiments could be performed: a surface reference expressed in the robot base frame, and the normal direction of the physical tool face expressed in the end-effector frame. Each came from its own calibration procedure.

The surface-plane calibration of Section 4.2.1 supplies a point p_s on the plate and its outward normal n_s, both in the base frame. The two tangent directions t_1 and t_2 are constructed from n_s, which completes the surface frame $R_{\text{surface}} = [\, t_1 \quad t_2 \quad n_s \,]$. The tool-normal calibration of Section 4.2.2 supplies $n_{\text{Tool,EE}}$, the direction normal to the physical tool face in end-effector coordinates. That second calibration is needed because the physical face is not exactly perpendicular to the nominal $+Z_{\text{EE}}$ axis.

Each symbol carries the frame it is expressed in. $n_{\text{Tool,EE}}$ is the tool-face normal in end-effector coordinates, and $n_{\text{Tool,0}}$ is the same physical direction in the base frame. Only the first is calibrated. The second follows from it during operation, as Section 3.2.2 sets out. It is used only for the pose-based consistency check reported in Appendix D.7.

Neither calibration entered the other. The surface reference was obtained first, from one seated pose, using the nominal $+Z_{\text{EE}}$ direction; the tool normal calibrated afterwards was not used to revise it. Determining $n_{\text{Tool,EE}}$ in turn used only the relative orientations of four seated yaw poses, and no value of n_s. That separation matters, because the calibrated $n_{\text{Tool,EE}}$ later differed from nominal $+Z_{\text{EE}}$ by approximately 1.56°.

Consequently the surface frame used by the controller is a configured geometric reference rather than an independent measurement of the physical plane normal. It was taken from the nominal end-effector axis at one seated pose. The 1.56° is the difference between those two orientation references, and it applies equally to every reported run. Since it limits an absolute interpretation of the physical tool–surface angle, the main comparisons report the signed end-effector set-up rotation of Section 4.5.1.

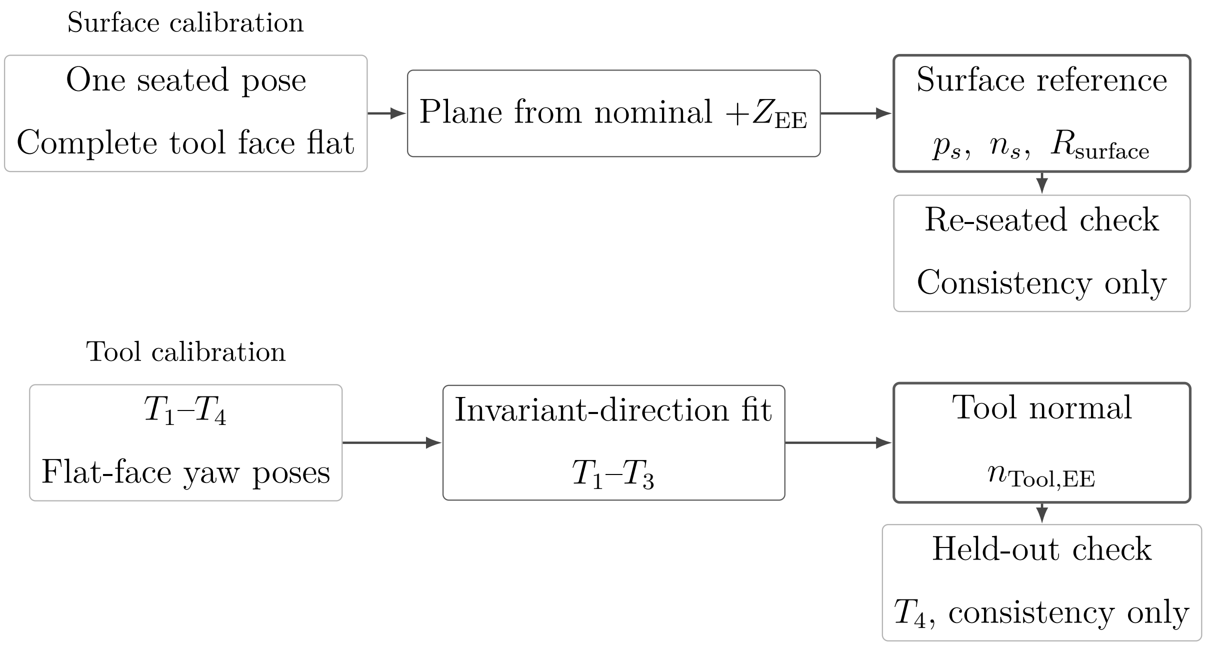


Figure 4.2: The two independent calibrations and the result each produces.

4.2.1 Surface-Plane Calibration

The complete tool face was seated on the plane and one end-effector pose was recorded. Its nominal $+Z_{\mathrm{EE}}$ direction supplied the surface normal n_s, and the tool-face centre supplied the plane point p_s. Projecting the base-frame x direction onto the plane gave t_1; $t_2 = n_s \times t_1$ completed $R_{\mathrm{surface}} = [t_1\ t_2\ n_s]$. The stored surface tilts were $-1.585°$ about base x and $+0.988°$ about base y. The calibrated values and their construction are collected in Appendix C.

4.2.2 Tool Normal Calibration

The complete tool face was seated at four yaw orientations. The direction in end-effector coordinates that remained most nearly invariant in the base frame was fitted from the first three captures; the fourth was retained as a consistency capture. The resulting tool normal $n_{\mathrm{Tool,EE}}$ differed from nominal $+Z_{\mathrm{EE}}$ by approximately $1.56°$. Its value and the minimisation used for the fit are given in Appendix C.

4.2.3 Calibrated Experimental Geometry

The rectangular face geometry used by the controller is listed in Table 4.1. The leading-point selection and the reconstruction of the TCP target are described in Section 3.2.3.

Table 4.1: Calibrated tool geometry.

Quantity	Value in end-effector coordinates
Tool-face centre	$[0, 0, 0.020]^\top$ m
Half-width	$[0.020, 0, 0]^\top$ m
Half-length	$[0, 0.060, 0]^\top$ m
Feature-selection tolerance	0.1 mm

4.3 Common Contact-Test Configuration and Procedure

All surface-contact cases used the same calibrated geometry, phase sequence, normal press trajectory, fixed gain entries, and null-space configuration except where a parameter was explicitly varied. This section defines that common experimental basis before the individual cases are introduced in Section 4.4.

4.3.1 Common Cartesian Gains and Damping

The phase-dependent gains are diagonal in the calibrated surface frame $[t_1, t_2, n_s]$ before any compliance-centre shift. Table 4.2 gives the entries used by every reported contact run, together with the damping ratio ζ of the corresponding phase. The configured grinding phase carried the set-up entries unchanged. The pre-set-up gate was configured with the same stiffness as set-up and was disabled in the reported runs, so it did not act as a separate phase.

Table 4.2: Common Cartesian gains and damping ratio by phase.

	K_p [N/m]			K_R [N m/rad]			
Phase	t_1	t_2	n_s	t_1	t_2	n_s	ζ
Orientation and approach	150	150	150	180	180	90	1.9
Set-up and grinding	2000	2000	350	5	5	50	1.0

During orientation and approach the translational stiffness was isotropic and the rotational stiffness was higher about the surface tangents. In set-up the lower normal entry limits the elastic normal-force increase for a given normal position error, while the higher tangential entries restrict motion in the surface plane. The low rotational stiffnesses about t_1 and t_2 permit contact-induced alignment rotation, and the higher entry about n_s mainly resists in-plane spin of the tool face. Because the set-up translational stiffness $K_{p,\text{set}}$ is diagonal in the surface frame, the effective normal stiffness $K_{p,n} = n_s^\top K_{p,\text{set}} n_s$ equals the tabulated normal entry of $350\,\text{N/m}$.

The directional damping coefficients were calculated from the Cartesian inertia and the active stiffness at the tabulated damping ratio, as described in Section 3.4.2. In contrast, the fallback damping values, used only if the inertia-based calculation could not be applied, are provided in Appendix C.

4.3.2 Contact Sequence and Set-Up Motion

The common trajectory and transition parameters are collected in Table 4.3. These settings were used throughout the surface-contact experiments unless explicitly stated otherwise.

Table 4.3: Common phase-transition and trajectory parameters for the surface-contact experiments.

Quantity	**Value**	**Role**
Minimum orientation time	1.0 s	Prevents an immediate phase transition.
Tool-axis transition tolerance	0.016 rad	Maximum remaining tool-axis error before descent.
Spin transition tolerance	0.009 rad	Maximum remaining normal-axis spin error.
Orientation timeout / maximum rate	8.0 s / 20°/s	Bounds the orientation phase.
Descent speed	0.1 m/s	Reference speed along the descent direction.
Maximum descent distance	0.640 m	Terminates an unsuccessful approach.
Surface clearance	0.020 m	Geometric transition from approach to set-up.
Minimum set-up time	2.0 s	Earliest activation of the moment-based exit condition.
Set-up timeout	5.0 s	Time-based set-up termination.

Table 4.3: Common phase-transition and trajectory parameters for the surface-contact experiments (continued).

Quantity	**Value**	**Role**
Moment-change threshold	150 N m	Alternative set-up termination based on the model-estimated external moment.
Set-up push speed	0.080 m/s	Rate of the signed press coordinate.
Set-up push endpoint	+0.240 m	Final signed Cartesian spring reference.
Pre-set-up gate	disabled	Approach transfers directly to set-up.
Pre-grind gate	enabled	Holds the reached pose before the optional grinding transition.
Configured grinding sweep	t_2, 0.030 m, 0.2 Hz	Implemented but not entered in the reported runs.

Before each run, the active parameter set was loaded and the robot was moved to the configured initial joint posture. The calibrated geometry, the phase gains and damping, and the compliance-centre definition were prepared before the torque callback started.

Approach ended when the selected tool point reached 20 mm above the calibrated plane. This transition was defined geometrically rather than by a force threshold. At the same instant, the controller stored the TCP pose, the orientation reference, the selected tool point, and its projection onto the plane. The model-estimated external wrench was additionally stored for the optional wrench-change termination condition described in Section 3.5.3.

The set-up press endpoint of +0.240 m is a Cartesian spring reference beyond the calibrated plane and is not interpreted as a physical penetration of the same magnitude. The reference reached it after approximately 3.25 s and was then held. Throughout set-up the rotational reference remained the orientation captured at the clearance transition, so the measured alignment rotation is a response to contact rather than the tracking of a time-varying orientation command. The TCP-centred and stiffness-variation conditions used the decoupled Cartesian impedance during set-up. Conditions with a displaced CoC used the point-shifted 6×6 impedance associated with the selected compliance-centre position.

Every reported contact run ended at the 5 s set-up timeout, and the moment-change threshold was reached in none of them. The pre-grind gate then held the reached pose. No run entered the configured grinding sweep, so the recorded data cover orientation, approach, and set-up only.

Fixed null-space configuration. Projected null-space damping and singular-value conditioning were active together throughout the surface-contact measurements. Keeping this secondary controller fixed prevents changes in the null-space policy from becoming an additional variable across the surface-contact experiments. The gains were $d_{\mathrm{null}} = 2\,\mathrm{N\,m\,s/rad}$ and $k_\sigma = 1\,\mathrm{N\,m}$.

This is the combined secondary-torque mode of the implementation described in Section 3.6. The contact measurements therefore evaluate the surface-alignment controller with both null-space terms active. Their individual effects are examined separately in the pose-hold experiment of Section 4.6.

4.3.3 Repetitions and Data Recording

The recorded signals and CSV structure are documented in Section 3.7.2 and appendix B.

Three repetitions were recorded for each parameter setting. Results are reported as arithmetic means, with sample standard deviations given for the main comparisons and where relevant in the supporting checks. The main surface-contact study comprises 111 runs across 37 settings. A further 54 runs across 18 settings were performed as supporting checks and are reported in Appendix D. In total, 165 surface-contact runs across 55 settings were recorded. No run was discarded. Together with the 12 pose-hold runs of Section 4.6, the complete experimental data set contains 177 recorded runs.

Within each controlled comparison, the calibrated geometry, common phase parameters, fixed gains, and null-space settings remained unchanged. Only the parameter assigned to the corresponding comparison was varied.

4.4 Main Surface-Contact Experimental Matrix

The main surface-contact study separates the effects of the principal controller parameters. Case A establishes the TCP-centred reference, where $r_c = 0$. In Cases B and C, rotational and translational stiffness were varied independently so that each gain could be evaluated without changing the other simultaneously. In Case D the tangential compliance-centre position was varied across both principal surface tangents and both signs of the commanded orientation offset.

Unless stated otherwise, the non-zero compliance-centre displacements of the main study were configured tool-fixed, in end-effector coordinates. The tangential coordinates reported for Case D are the surface-frame directions those settings realise in the flat target orientation. The definition frame itself is examined in the supporting check of Appendix D.3.

The orientation-offset-magnitude comparison, the CoC definition frame, displacement along the tool axis, and intermediate tangent-plane directions are treated as supporting checks in Appendix D.

Table 4.4 summarises the parameter variations of the main study. Each listed setting was repeated three times unless it is a reference condition counted with an earlier case.

Table 4.4: Main surface-contact cases and parameter variations.

Case	Varied quantity	Commanded tool orientation offset	Tested values	Runs	Evaluated effect
A	CoC at the TCP	None and $\pm 10°$ about t_1 or t_2	$r_c = 0$.	15	Contact response without CoC coupling.
B	Rotational stiffness	$+10°$ about t_1 or t_2	Command-axis $K_R = 15$ and 50 N m/rad.	12	Influence of rotational stiffness.
C	Translational stiffness	$+10°$ about t_1 or t_2	Cross-axis $K_p = 300$ and 800 N/m.	12	Influence of translational stiffness.
D	Tangential CoC position	$\pm 10°$ about t_1 or t_2	-40, -20, -10, $+10$, $+20$, $+40$ mm along the tangent perpendicular to the commanded rotation.	72	Direction and sign dependence.

The run counts in Table 4.4 give the three scheduled repetitions of each setting. A condition that a case shares with an earlier one is counted with that earlier case.

In Cases B and C, only one Cartesian stiffness entry was varied within each comparison. The remaining gain entries were kept at the common values of Table 4.2.

4.5 Evaluation Quantities

The primary response quantity for the surface-contact experiments is the signed end-effector rotation during set-up. It measures how much the end effector rotated after the orientation reference was fixed at the clearance transition. The commanded Cartesian wrench is retained for the Case-D mechanism plot.

4.5.1 Signed End-Effector Set-Up Rotation

Set-up holds the orientation captured at the clearance transition as its rotational reference, so the logged base-frame orientation error e_R is the rotation from the current end-effector orientation back to that reference. Since that reference is captured at the beginning of set-up, $e_R = \mathbf{0}$ at set-up entry. Its magnitude at the end of set-up therefore gives the angular change relative to the held reference. The reported rotation vector is

$$\Delta\theta = e_R\Big|_{\text{end of set-up}}. \tag{4.1}$$

This quantity is obtained from the measured end-effector orientation alone, and the calibrated tool normal does not enter it. It is therefore independent of the relative rotation between the tool and the gripper described in Section 4.1.1, which was not tracked during the contact experiments. The surface frame enters solely as the directions the rotation is resolved along:

$$\Delta\theta^S = R_{\text{surface}}^{\top}\Delta\theta, \qquad \Delta\theta_i = t_i^{\top}\Delta\theta, \quad i \in \{1,2\}. \tag{4.2}$$

The reported quantity $\Delta\theta_i$ is the signed end-effector set-up rotation about surface tangent t_i. Because e_R is the rotation from the measured end-effector orientation back to the held reference, its sign is opposite to that of the physical end-effector motion. A positive $\Delta\theta_i$ therefore denotes alignment-directed rotation for a positive commanded offset. A negative value denotes rotation in the opposite direction, and reversing the command reverses the sign of the intended direction. The measurements confirm the correspondence: for the $+10^\circ$ command about t_1 at the TCP, $\Delta\theta_1 = +7.57^\circ$ accompanies a fall in the pose-based alignment angle from 9.32° to 3.38°, whereas the $+10^\circ$ command about t_2 gives $\Delta\theta_2 = -1.63^\circ$ with that angle rising from 10.07° to 11.81°. This quantity is used for all main comparisons in Chapter 5.

4.5.2 Supporting Geometric Checks

A separate pose-based alignment angle calculated from the calibrated tool normal is retained only as a supporting consistency check in Appendix D.7. It is not used for the main comparisons because it is unsigned and depends on an assumed fixed relation between the tool and end effector.

A separate geometry-based flatness classification is obtained from the final TCP height relative to the calibrated plane and the height expected from the calibrated rectangular tool geometry when the tool face is flat. This classification uses the end-effector position and calibrated geometry; it is not a direct angular measurement of the physical tool face. The TCP-height classification is therefore reported separately from the primary end-effector set-up rotation.

4.5.3 Commanded Wrench Quantities

The commanded normal force and commanded TCP moment used in the controller-response plots are obtained from $F = [f^\top, m^\top]^\top$:

$$F_n = n_s^\top f, \qquad M_{t_i} = t_i^\top m. \tag{4.3}$$

F_n is signed, and it is negative while the tool presses into the surface. Both quantities are controller commands, and the plotted moment is therefore the complete TCP moment generated by the point-shifted spring–damper law.

4.6 Secondary Null-Space Pose-Hold Experiment

The surface-contact measurements keep projected damping and singular-value conditioning active together. A separate Cartesian pose-hold experiment was therefore used to compare their individual effects under a repeatable internally commanded disturbance.

4.6.1 Experimental Configuration

The pose-hold controller maintained the initial TCP pose with the gains of Table 4.5. Every entry was isotropic, so each matrix is the stated value times the identity. Unlike the contact-sequence phases, the twelve pose-hold trials used fixed Cartesian damping rather than inertia-scaled damping.

Table 4.5: Isotropic Cartesian gains of the pose-hold trials.

Quantity		**Value**
Translational stiffness	$K_{p,\mathrm{hold}}$	2000 N/m
Rotational stiffness	$K_{R,\mathrm{hold}}$	50 N m/rad
Translational damping	$D_{p,\mathrm{hold}}$	50 N s/m
Rotational damping	$D_{R,\mathrm{hold}}$	8 N m s/rad

Four null-space settings were tested with three repetitions each: no null-space torque, projected damping with $d_{\mathrm{null}} = 2\,\mathrm{N\,m\,s/rad}$, sigma conditioning with $k_\sigma = 1.5\,\mathrm{N\,m}$, and sigma conditioning with $k_\sigma = 2.0\,\mathrm{N\,m}$. The probe step, sigma deadband, and relative SVD threshold remained at 0.08 rad, 10^{-6}, and 10^{-4}, respectively.

4.6.2 Commanded Disturbance

The disturbance represents the joint-torque equivalent of a commanded point force acting at a specified location on link 3. The point was defined by the link-frame offset

$$r_p = \begin{bmatrix} 0 \\ 0 \\ 0.200 \end{bmatrix} \mathrm{m}. \tag{4.4}$$

At the initial posture q_{init}, the structural null direction $v_7(q_{\text{init}})$ and the translational point Jacobian $J_p(q_{\text{init}})$ define the fixed base-frame force direction

$$u_f = \frac{J_p(q_{\text{init}})v_7(q_{\text{init}})}{\|J_p(q_{\text{init}})v_7(q_{\text{init}})\|_2}. \tag{4.5}$$

The commanded force magnitude was $F_d = 20\,\mathrm{N}$. The virtual force and its equivalent joint torque were

$$f_d(t) = F_d s_d(t) u_f, \qquad \tau_{\text{dist}}(t) = J_p(q(t))^\top f_d(t). \tag{4.6}$$

Here u_f is the fixed unit force direction defined in Equation (4.5), and $s_d(t) \in [0, 1]$ is a dimensionless time-scaling function that switches the commanded virtual disturbance smoothly on and off. It was defined as

$$s_d(t) = \begin{cases} 0, & t < 5\,\mathrm{s}, \\ \frac{1}{2}\left[1 - \cos\left(\pi \frac{t - 5\,\mathrm{s}}{2\,\mathrm{s}}\right)\right], & 5\,\mathrm{s} \le t < 7\,\mathrm{s}, \\ 1, & 7\,\mathrm{s} \le t < 8\,\mathrm{s}, \\ \frac{1}{2}\left[1 + \cos\left(\pi \frac{t - 8\,\mathrm{s}}{1\,\mathrm{s}}\right)\right], & 8\,\mathrm{s} \le t < 9\,\mathrm{s}, \\ 0, & t \ge 9\,\mathrm{s}. \end{cases} \tag{4.7}$$

Thus the disturbance increased smoothly from zero to the full commanded magnitude between 5 and 7 s, remained at full magnitude until 8 s, and decreased smoothly to zero by 9 s. The equivalent disturbance torque was limited to

$$\|\tau_{\text{dist}}\|_2 \le 3\,\mathrm{N\,m}.$$

The disturbance is internal to the controller and therefore provides a repeatable comparison input; it is not a separately applied physical force.

In every trial the selected null-space law was active from the start of the pose-hold phase. The first 5 s therefore formed a pre-disturbance settling interval, during which the conditioning torque of the sigma-only settings could already move the redundant configuration.

The disturbance then acted from 5 to 9 s, followed by an observation interval until the end of the trial. No secondary term was enabled or disabled within a trial: the sigma-conditioning torque remained active throughout the complete sigma-only trial and was not switched on after the disturbance had been removed.

4.6.3 Evaluation Quantities

All quantities in this subsection are evaluated over the interval in which the internally commanded virtual disturbance acts, $t \in [5, 9]\,\mathrm{s}$. The first criterion checks whether the primary Cartesian pose-hold task remains approximately unchanged while the disturbance and the selected null-space law act. The predefined task-retention criterion is

$$\max_{5\,\mathrm{s} \le t \le 9\,\mathrm{s}} \|e_p(t)\|_2 < 2\,\mathrm{mm}, \tag{4.8}$$

where $e_p(t)$ is the Cartesian position-error vector. A trial therefore satisfies the criterion when the TCP position-error norm remains below 2 mm throughout the commanded disturbance interval.

The redundant joint motion is evaluated from the projected joint velocity

$$\dot{q}_{\mathrm{null}}(t) = N_q(q(t))\dot{q}(t), \tag{4.9}$$

where $N_q(q)$ is the kinematic null-space projector. The vector $\dot{q}_{\mathrm{null}} \in \mathbb{R}^7$ has units of rad/s and represents the component of the measured joint velocity that lies in the instantaneous kinematic null space.

Two complementary quantities are formed from this projected velocity. The first is the cumulative projected null-space motion

$$E_N = \int_{5\,\mathrm{s}}^{9\,\mathrm{s}} \|\dot{q}_{\mathrm{null}}(t)\|_2 \,\mathrm{d}t. \tag{4.10}$$

Because the norm is taken before integration, motion in opposite directions contributes positively in both cases. E_N therefore measures the total amount of redundant joint motion that occurred during the commanded disturbance interval. Its unit is radians and the reported values are converted to degrees. For example, motion of $+5°$ along a redundant direction followed by $-5°$ back along the same direction contributes approximately $10°$ to E_N, even though the final redundant configuration can be close to its starting value. In this sense E_N is a joint-space path-length quantity, not a torque or energy measure.

The second quantity retains the direction of the motion. Integrating the projected velocity without taking its norm gives the accumulated projected joint displacement

$$\Delta q_{\text{null}} = \int_{5\,\text{s}}^{9\,\text{s}} \dot{q}_{\text{null}}(t)\,\mathrm{d}t = \int_{5\,\text{s}}^{9\,\text{s}} N_q(q(t))\dot{q}(t)\,\mathrm{d}t. \tag{4.11}$$

The vector $\Delta q_{\text{null}} \in \mathbb{R}^7$ has units of radians. It describes the net accumulated projected joint motion during the disturbance interval, so motion in opposite directions can cancel. It is not an integrated torque: the integrated quantity is joint velocity.

A common direction is introduced to express this seven-dimensional net displacement by one signed scalar. Let $\Delta q_{\text{null},0}$ denote the reference net projected displacement obtained from the condition without null-space torque. The corresponding unit direction is

$$v_{\text{ref}} = \frac{\Delta q_{\text{null},0}}{\|\Delta q_{\text{null},0}\|_2}. \tag{4.12}$$

Thus v_{ref} represents the redundant joint-space direction in which the commanded virtual disturbance moves the robot in the reference condition without a secondary null-space torque. The same direction is used for all tested null-space settings. Their signed net displacement along this common direction is

$$\Delta\eta_{\text{dist}} = v_{\text{ref}}^{\top}\Delta q_{\text{null}}. \tag{4.13}$$

A positive $\Delta\eta_{\text{dist}}$ denotes net motion in the same direction as the reference disturbance response, a negative value denotes net motion in the opposite direction, and a value near zero indicates little remaining net displacement along that direction. The quantity is a signed projection of the accumulated projected joint motion rather than an exact nonlinear redundant-coordinate displacement, because $N_q(q)$ changes with the robot configuration while v_{ref} is held fixed. At full row rank the Panda has a one-dimensional instantaneous null space, so this common direction provides a meaningful signed comparison of the four tested settings.

The two redundant-motion metrics therefore answer different questions: E_N measures how much redundant joint motion occurred in total, whereas $\Delta\eta_{\mathrm{dist}}$ measures how much net displacement remained along the common reference disturbance direction. A trajectory can consequently have a comparatively large E_N while ending close to its starting redundant configuration and therefore having a small $\Delta\eta_{\mathrm{dist}}$.

Singular-value conditioning was evaluated from the change in $\sigma_{\min}$ across the same disturbance interval, and Cartesian task retention from the peak position-error norm. The corresponding results are presented in Chapter 5.

Chapter 5

Results and Discussion

The surface-contact experiments evaluate how the controller parameters influence the end-effector rotation generated during contact set-up. The primary response quantity is the signed set-up rotation $\Delta\theta_i$ about the investigated surface tangent t_i. It is measured from the beginning to the end of set-up, and its sign follows the surface-frame rotation convention of Section 4.5.

The commanded tool orientation offset θ_{t_i} defines the initial experimental condition, whereas the measured response is the signed end-effector set-up rotation $\Delta\theta_i$. The main response figures therefore report $\Delta\theta_i$, while the corresponding numerical values are provided in Appendix D. The pose-based alignment estimate and the TCP-height flatness classification are retained only as supporting checks in the same appendix.

Each parameter setting was repeated three times. The response figures show the mean over the repetitions, and the Case-D comparison additionally carries error bars. The sample standard deviation of every reported setting is given in Appendix D. The preceding orientation phase terminated with a finite tracking tolerance, so the initial angular magnitude differed slightly from the commanded value. A commanded $+10°$ offset reached approximately $9.3°$ about t_1 and $10.0°$ about t_2. The commanded offset identifies each experimental condition.

In the response figures that follow, a positive $\Delta\theta_i$ denotes rotation in the alignment direction for a positive commanded offset. A negative value denotes rotation in the opposite direction.

5.1 Main Surface-Contact Results

The cases are reported in the order in which they constrain the choice of compliance-centre position. The TCP-centred reference is reported in Case A, followed by the stiffness sensitivities in Cases B and C. In Case D the tangential compliance-centre displacement was tested over both surface tangents and both signs of the commanded orientation offset. The orientation-magnitude, definition-frame, tool-axis, and intermediate-direction comparisons are reported as supporting checks in Appendix D.

5.1.1 Baseline Contact Response with the CoC at the TCP: Case A

In Case A the CoC was placed at the TCP, so $r_c = 0$. This condition removes the additional translation–rotation coupling introduced by the virtual point shift. The ordinary rotational impedance and physical contact geometry remain. The measured response therefore provides the baseline for the stiffness and compliance-centre-position comparisons.

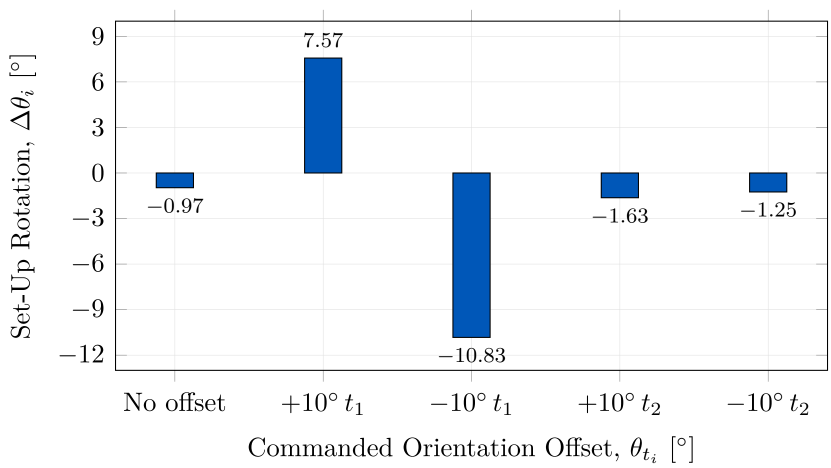


Figure 5.1: Measured set-up rotation with the CoC at the TCP.

With the CoC at the TCP, the measured response differs strongly between the two surface tangents. For the positive 10° conditions, Figure 5.1 gives +7.57° about t_1 and −1.63° about t_2. Reversing the commanded orientation changes the response to −10.83° and −1.25°, respectively. The figure shows all five baseline conditions.

The condition without a commanded orientation offset gave −0.97° about t_1. Contact produced end-effector rotation from the flat target orientation as well. The commanded conditions are read against that non-zero response rather than against zero. The TCP-centred condition provides the reference against which the stiffness and compliance-centre-position variations are compared.

5.1.2 Effect of Cartesian Stiffness: Cases B and C

Rotational Stiffness: Case B

In Case B the rotational stiffness about the investigated tangent was varied while the other controller settings remained unchanged. The 5 N m/rad condition is the corresponding TCP-centred reference from Case A.

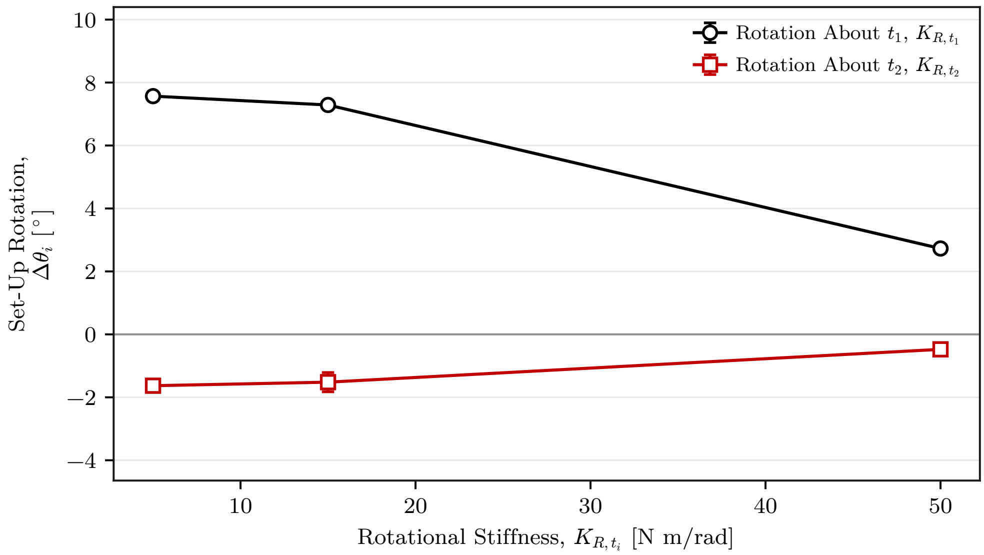


Figure 5.2: Measured set-up rotation by rotational stiffness.

Increasing K_{R,t_i} reduced the measured rotation over the tested range. About t_1, Figure 5.2 shows a decrease from $+7.57°$ at 5 N m/rad to $+2.73°$ at 50 N m/rad. About t_2, the response remained small and changed from $-1.63°$ to $-0.48°$. Rotational stiffness therefore changed the response magnitude without removing the directional difference observed in Case A.

Cross-Axis Translational Stiffness: Case C

In Case C the translational stiffness perpendicular to the investigated rotation axis was varied. The rotational stiffness about that axis was held at 5 N m/rad. The 2000 N/m condition is the corresponding TCP-centred reference from Case A.

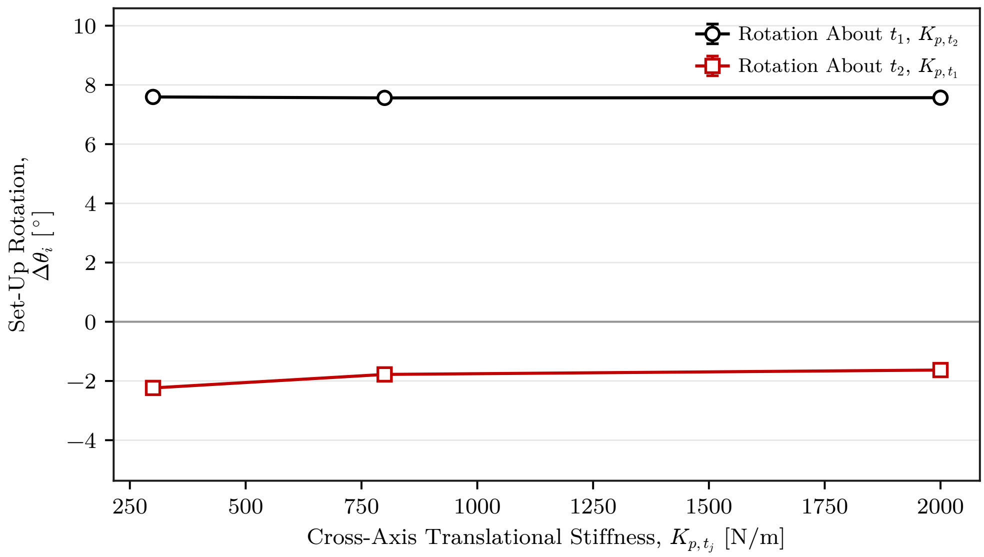


Figure 5.3: Measured set-up rotation by cross-axis translational stiffness.

The influence of perpendicular translational stiffness was considerably smaller than that of rotational stiffness. About t_1, the measured rotation remained within 0.03° across the tested range. About t_2, Figure 5.3 shows a change of 0.61°, from −2.24° to −1.63°. Varying this stiffness alone therefore did not account for the directional difference in the baseline response.

5.1.3 Effect of Tangential CoC Position: Case D

In Case D the tangential position of the CoC was varied while the Cartesian gains remained unchanged. For each rotation axis, the centre was moved along the surface tangent perpendicular to that axis. The TCP-centred condition is included as the zero-position reference.

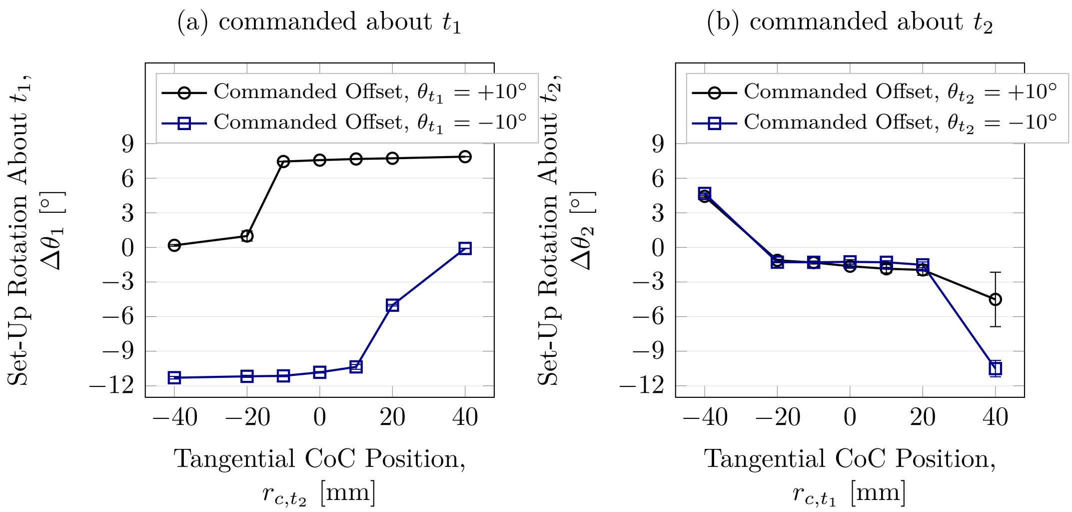


Figure 5.4: Set-up rotation by tangential CoC position (mean ± SD).

Tangential CoC position produced the largest response variation among the tested controller parameters. As shown in Figure 5.4, reversing the initial orientation condition changes which side of the TCP produces the larger response. About t_1, the positive condition gives $+7.87°$ at $+40$ mm and $+0.18°$ at -40 mm. The reversed condition exchanges the larger response between the two sides.

Changing the rotation axis also changes the required tangential direction. For a positive condition about t_2, the response changes from $-1.63°$ at the TCP to $+4.43°$ at the selected $r_{c,t_1} = -40$ mm position. The selected displacement therefore lies along $+t_2$ for rotation about t_1 and along $-t_1$ for rotation about t_2. Within the investigated configuration, a non-zero tangential CoC therefore cannot be selected independently of the direction and sign of the initial angular condition.

The error bars in Figure 5.4 show that two transition conditions had larger spread than the other lever positions. The positive t_1 condition at -20 mm had a standard deviation of $0.44°$. The positive t_2 condition at $+40$ mm had a standard deviation of $2.36°$.

The positive t_1 condition was examined in the time domain to relate the rotation response to the commanded wrench. Its two outer positions produce a large difference in $\Delta\theta_1$, while the TCP-centred condition provides the zero-position reference. Each trace is one recorded repetition of its centre position.

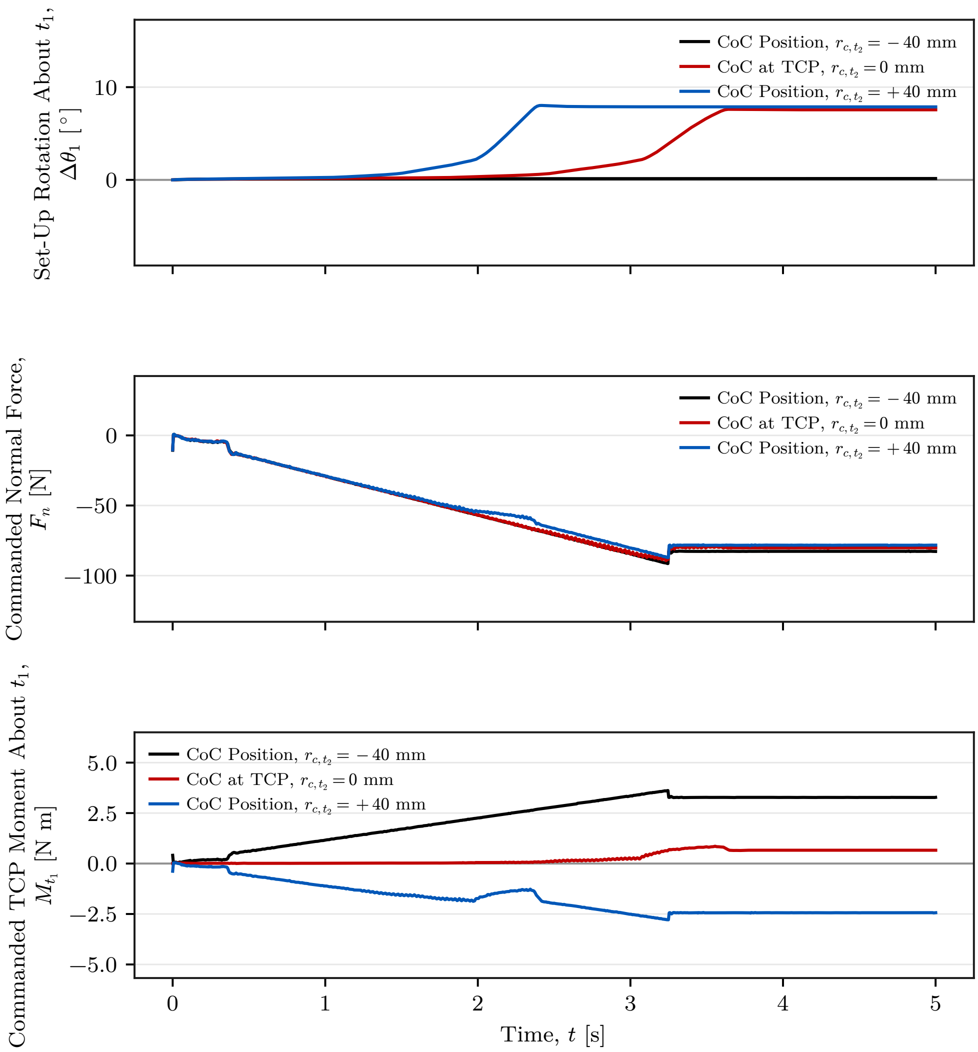


Figure 5.5: Measured set-up rotation, commanded normal force, and commanded TCP moment for three tangential CoC positions.

Figure 5.5 places the set-up rotation $\Delta\theta_1$, the commanded normal force F_n, and the commanded TCP moment M_{t_1} on a common time axis for three centre positions. Every condition shown carries a commanded orientation offset of $+10°$ about t_1. At -40, 0, and $+40$ mm, the commanded normal force settles in a similar range of approximately -77 to -82 N, while the commanded TCP moment about t_1 settles at approximately $+3.28$, $+0.66$, and -2.44 N m. The outer positions therefore produce moments of similar magnitude and opposite sign while their set-up rotations differ by 7.73°. The difference in set-up rotation is therefore associated primarily with the moment generated by the displaced impedance reference rather than with a substantial change in the normal pressing force.

The commanded moment and the reported rotation carry opposite signs here. $\Delta\theta_1$ is the rotation from the measured end-effector orientation back to the reference held through set-up, as Section 4.5.1 defines it. Its sign is therefore opposite to that of the physical rotation. The alignment-directed response at +40 mm appears as a positive $\Delta\theta_1$ against a negative commanded M_{t_1}.

A tangential displacement of the CoC moves the impedance reference point away from the TCP, so the translational press is coupled to the rotational part of the impedance. The complete commanded moment is $m = m_R + r_c \times f$, with $r_c = p_c - p_{\mathrm{TCP}}$. A non-zero tangential displacement therefore adds a moment contribution from the commanded press. This explains why the commanded moment changes strongly with centre position while F_n remains within a similar range. Reversing the tangential displacement reverses the contribution $r_c \times f_n$ generated by the normal press.

The measured sign reversal follows the relation derived in Section 2.7.2. A fixed tangential $r_c \neq 0$ retains one preferred moment direction while the press continues. At $r_c = 0$, the point-shift contribution vanishes and the TCP selects no tangential direction. The TCP-centred condition is therefore the direction-independent fixed reference among the tested positions. A displaced centre provides greater condition-specific rotational authority where the required direction is known.

The grinding phase returns to the decoupled impedance branch described in Section 3.5.4. This phase structure is consistent with removing the direction-selective coupling after set-up. It is an implementation property rather than an experimental result, because the reported runs ended at the pre-grinding gate.

Across Cases A–D, tangential CoC position produced the largest measured response variation among the tested controller parameters.

5.2 Null-Space Pose-Hold Results

The remaining experiment is separate from the contact cases and analyses the redundant joint-space direction while the Cartesian pose is held. Matched trials compare the selectable null-space terms under the same internally commanded point-force-equivalent disturbance. The commanded disturbance reached 20.00 N in all twelve trials, with peak equivalent joint-torque norms between 2.17 and 2.45 N m. The torque-limit scale remained equal to one, so the waveform was applied without clipping.

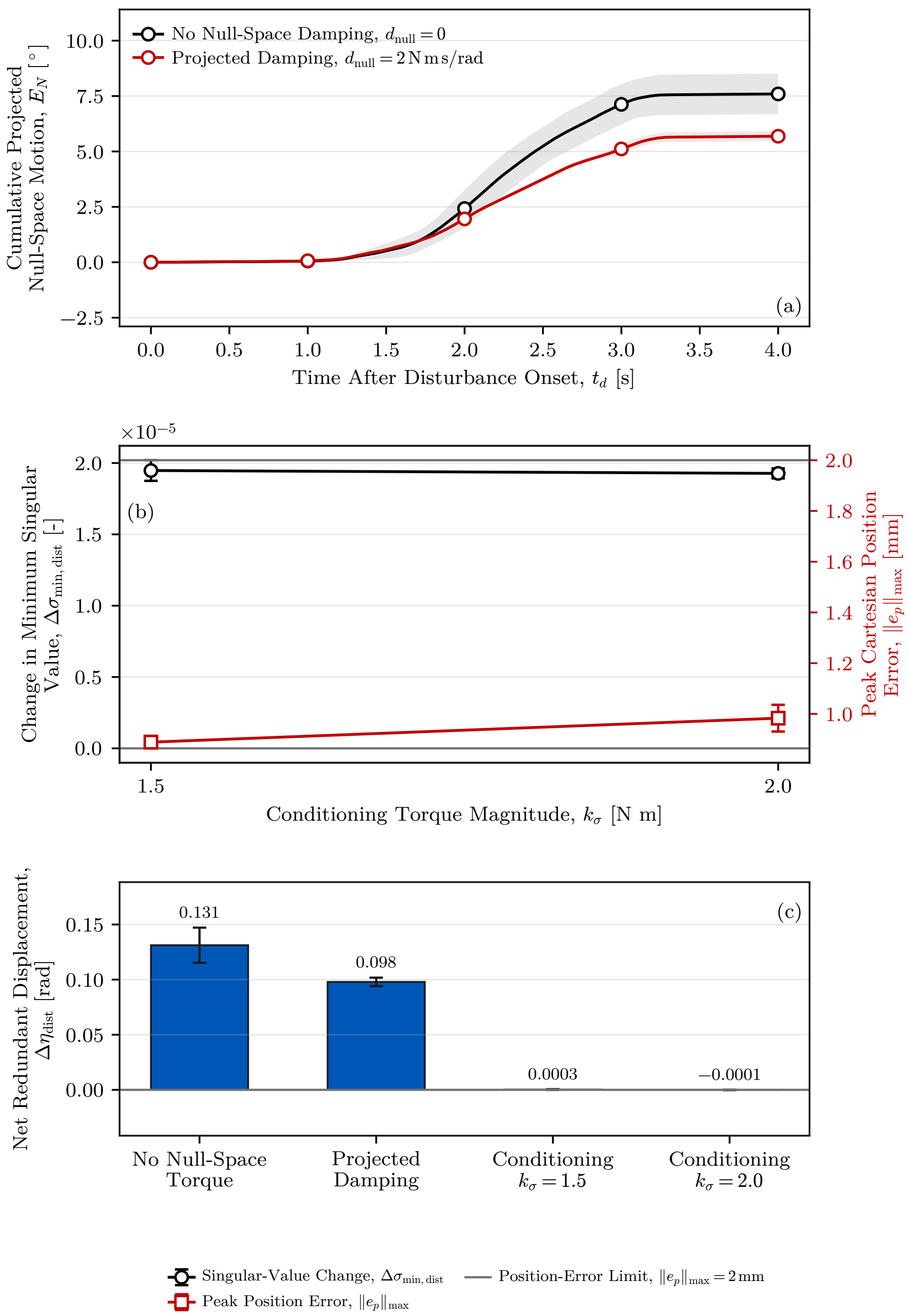


Figure 5.6: Null-space pose-hold response.

Figure 5.6 reports the response during the Cartesian pose-hold disturbance test in three panels. Panel (a) gives the mean cumulative projected null-space motion E_N without null-space torque and with projected damping, the shaded bands showing ± one standard deviation across the three repetitions. Panel (b) gives the change in the minimum Jacobian singular value over the disturbance interval, $\Delta\sigma_{\text{min,dist}}$, together with the peak Cartesian position error, for the two singular-value conditioning magnitudes; its error bars show $\pm$ one standard deviation and the horizontal line marks the 2 mm task-retention limit. Panel (c) gives the net redundant displacement $\Delta\eta_{\text{dist}}$ of all four settings, with the mean printed above each bar and error bars showing $\pm$ one standard deviation. The complete numerical values are given below.

5.2.1 Projected Null-Space Damping

Projected null-space damping reduced E_N from $(7.596 \pm 0.916)^\circ$ without null-space torque to $(5.687 \pm 0.228)^\circ$ at $d_{\text{null}} = 2\,\text{N}\,\text{m}\,\text{s/rad}$, a reduction of approximately 25 %. The net redundant displacement fell in the same proportion, from $0.131 \pm 0.016\,\text{rad}$ to $0.098 \pm 0.004\,\text{rad}$, and the change in σ_{min} across the disturbance interval became less negative, from $(-2.13 \pm 0.47) \times 10^{-3}$ to $(-1.26 \pm 0.10) \times 10^{-3}$. Damping acts on the projected joint velocity through $\tau_d = -d_{\text{null}} N_\tau \dot{q}$, so it vanishes as the redundant motion stops and requests no return towards the initial configuration.

5.2.2 Singular-Value Conditioning

Singular-value conditioning behaved differently from both the uncontrolled and the damping-only settings. It was already active during the 5 s preceding the disturbance and remained active throughout it, so it did not act as a term applied after the redundant configuration had been displaced. Both sigma-only settings ended the disturbance interval with a net redundant displacement whose magnitude was smaller than the scatter across the three repetitions: $(0.3 \pm 0.3) \times 10^{-3}\,\text{rad}$ at $k_\sigma = 1.5\,\text{N}\,\text{m}$ and $(-0.1 \pm 0.2) \times 10^{-3}\,\text{rad}$ at $2.0\,\text{N}\,\text{m}$. As panel (c) of Figure 5.6 shows, both are smaller than the 0.131 rad measured without null-space torque by more than two orders of magnitude. Over the same interval σ_{min} changed by $(1.95 \pm 0.07) \times 10^{-5}$ and $(1.93 \pm 0.03) \times 10^{-5}$, where it fell by $(-2.13 \pm 0.47) \times 10^{-3}$ without null-space torque. Under the tested disturbance direction, the conditioning term acted concurrently with the commanded disturbance and strongly suppressed the resulting redundant displacement.

The two sigma magnitudes differed in motion activity rather than in net displacement. At $k_\sigma = 2.0\,\text{N}\,\text{m}$ the cumulative projected null-space motion was $(1.619 \pm 0.130)^\circ$, against

$(0.283 \pm 0.007)^\circ$ at 1.5 N m, a factor of about six, while both net displacements stayed near zero. The larger torque magnitude was associated with greater switching activity near the locally preferred region. This is consistent with the form of the law: the conditioning term commands a fixed torque magnitude along the selected direction, and the selector s_σ is set to zero only inside the comparison deadband. On that reading the larger cumulative motion is switching activity around the same region rather than poorer disturbance rejection. At 1.5 N m the mean peak Cartesian position error was also lower, 0.889 ± 0.024 mm against 0.983 ± 0.053 mm. The largest position error measured in the complete pose-hold study was 1.304 mm, below the predefined 2 mm acceptance limit. Of the two tested magnitudes, 1.5 N m achieved the same suppression of net disturbance-induced displacement with substantially less redundant motion.

5.2.3 Interpretation and Experimental Bounds

The pose-hold trials compare the four selectable null-space modes under the repeatable commanded disturbance. The two terms act on different quantities. Projected damping dissipates redundant joint motion that is already present, whereas the singular-value term drives the redundant configuration towards the sampled neighbour with the larger $\sigma_{\min}$ whether or not such motion is present. The conditioning law is not dissipative: a disturbance acting towards larger $\sigma_{\min}$ would move the configuration in the same direction as τ_σ rather than against it, so the rejection observed here is specific to the tested disturbance direction. Both terms act through the null-space projector, so the Cartesian task stays primary.

Several bounds apply to these results. Because the conditioning torque was active before the disturbance, the sigma-only trials entered it from a redundant configuration displaced by approximately 0.013 rad from the one at which the trials without conditioning began. The four settings therefore compare the closed-loop behaviour of the complete modes. They do not isolate an instantaneous opposing torque at identical joint configurations. The surface-contact experiments use both contributions together in the combined secondary-torque mode and therefore do not separate their individual effects during contact. The geometric Jacobian combines translational and rotational rows without a common length scale, so the reported $\sigma_{\min}$ is used only as a relative indicator under the fixed Jacobian convention.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

A real-time Cartesian impedance controller was implemented for contact-induced alignment of a rectangular tool with a calibrated plane. The controller assigns directional gains in the surface frame and can shift the virtual centre of compliance from the TCP. The shift couples the commanded press to the rotational impedance through $m = m_R + r_c \times f$.

The TCP-centred conditions established a direction-dependent contact response. Substantial alignment-directed end-effector rotation was measured about t_1, while the response about t_2 was smaller and had the opposite sign for the positive command. Raising the rotational stiffness reduced the response over the tested range. Changing the cross-axis translational stiffness produced a smaller effect.

Tangential compliance-centre position produced the largest measured variation. The displacement that increased alignment depended on the sign of the initial orientation and on the commanded rotation axis. Reversing the displacement reversed the moment contribution from the normal press. Within the investigated task and tested positions, a non-zero tangential centre could therefore not be selected independently of the initial misalignment direction.

The TCP-centred impedance is the direction-independent fixed reference among the tested positions because it selects no tangential direction. A displaced centre provides greater condition-specific rotational authority when the required direction is known. These properties are distinct: the fixed reference need not produce the largest alignment-directed rotation in every condition.

The null-space pose-hold study separated projected joint damping from singular-value conditioning. Damping reduced redundant joint motion while the disturbance acted. Under the tested disturbance direction, conditioning strongly suppressed the net redundant displacement and kept the Cartesian position error below the specified 2 mm limit. The lower tested conditioning magnitude produced the same suppression with less cumulative redundant motion.

6.2 Limitations

6.2.1 Experimental Range

The contact study used one Panda configuration, one tool and mounting arrangement, one calibrated plane, and one press trajectory. The conclusions are therefore limited to this geometry and the tested stiffness, orientation-offset, and compliance-centre ranges. Sustained grinding and a centre scheduled during contact were not evaluated.

6.2.2 Measurement and Tool-Mount Uncertainty

The primary response was calculated from the measured end-effector rotation. The tool mount exhibited approximately $\pm 2°$ of unloaded play about y_{EE}, and its instantaneous motion was not tracked during contact. The results therefore describe end-effector rotation and do not provide an independent physical tool-face angle.

6.2.3 Null-Space Configuration and Real-Time Operation

The null-space terms were separated only in Cartesian pose hold under one internally commanded disturbance direction. The conditioning torque was active before the disturbance, so the experiment compares complete closed-loop modes rather than isolated torque contributions at identical configurations. The surface-contact study held the combined mode fixed. Worst-case execution time and scheduling jitter were not measured separately from operation through the nominal 1 kHz interface.

6.3 Future Work

The first priority is an independent measurement of the physical tool-face angle during contact. This would separate end-effector rotation from motion in the tool mount and provide a direct alignment reference.

The second priority is a direction-selected compliance-centre strategy. Such a controller would begin at the TCP, estimate the required alignment direction, apply a bounded tangential displacement during set-up, and return to the TCP before sustained contact. Negative orientation offsets and additional tangent-plane directions should be included in its evaluation.

The third priority is a combined-mode null-space study under physical disturbance. Varying projected damping and conditioning together would show whether the pose-hold findings remain valid during contact. A continuous or hysteretic conditioning law should also be compared with the present two-point selector to reduce its switching activity.

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Appendix A

Controller Source Listings

ORANGE — sufficient. The listings are abridged to the mathematical paths used by the thesis and connect each important equation to an implementation excerpt.

This appendix retains the two source paths that are non-obvious from the controller equations: the compliance-centre point shift and the final torque assembly. The excerpts use fixed-size Eigen aliases such as `Vec3`, `Vec7`, `Mat3`, `Mat6x6`, and `Mat6x7`. The identifier `R_base_surface` denotes the surface-frame orientation used in the chapters.

A.1 Centre of Compliance and the Offset Adjoint

The lever $r_c = p_c - p_{\mathrm{TCP}}$ points from the TCP to the selected centre of compliance, and the implementation uses the same convention. The same congruence is applied to stiffness and damping.

Listing A.1: TCP-to-compliance-centre adjoint and congruence transform.

```
Mat3 skewMatrix(const Vec3& v) {
  Mat3 s;
  s << 0.0, -v(2),  v(1),
       v(2),  0.0, -v(0),
      -v(1),  v(0),  0.0;
  return s;
}

Mat6x6 complianceShiftMatrix(const Vec3& r_c) {
  Mat6x6 adjoint = Mat6x6::Zero();
  adjoint.block<3, 3>(0, 0) = Mat3::Identity();
  adjoint.block<3, 3>(0, 3) = -skewMatrix(r_c);
  adjoint.block<3, 3>(3, 3) = Mat3::Identity();
  return adjoint;
}

Mat6x6 shiftGainToTcp(
    const Mat6x6& gain_at_center, const Vec3& r_c) {
  const Mat6x6 adjoint = complianceShiftMatrix(r_c);
  return adjoint.transpose() * gain_at_center * adjoint;
```

```
}
```

The wrench routine selects between the decoupled and the coupled law and constructs the lever. The centre can be defined relative to the TCP in end-effector coordinates, or the lever can be given directly in surface coordinates. Both definitions store r_c itself, so the two branches differ only in the frame it is expressed in.

Listing A.2: Selection of the decoupled or coupled spring wrench (abridged).

```
const bool coupled =
    params.use_virtual_compliance_center &&
    (phase == ControlPhase::kSetup ||
     (phase == ControlPhase::kPoseHold &&
      params.use_setup_impedance_hold));
if (!coupled) {
  // Two independent springs. dv.tail is -omega, so the damping
  // signs match.
  Vec6 wrench;
  wrench.head<3>() = Kp * dx.head<3>() + Dp * dv.head<3>();
  wrench.tail<3>() = KR * dx.tail<3>() + DR * dv.tail<3>();
  return wrench;
}

Vec3 r_c = Vec3::Zero();
if (params.compliance_center_in_tool_frame) {
  r_c = R_EE * params.compliance_center_offset_ee;
} else {
  r_c = R_base_surface * params.compliance_lever_surface;
}
const Mat6x6 K_tcp =
    shiftGainToTcp(blockDiagonal(Kp, KR), r_c);
const Mat6x6 D_tcp =
    shiftGainToTcp(blockDiagonal(Dp, DR), r_c);
return K_tcp * dx + D_tcp * dv;
```

A.2 Core Control Callback

The following excerpt shows the state and model quantities acquired at the start of each torque callback. The remaining listing then shows how these quantities enter the torque command.

Listing A.3: Robot-state and model quantities acquired by the torque callback.

```
Map<const Vec7> q_current(state.q.data());
Map<const Vec7> dq(state.dq.data());

const auto jacobian_array =
    model.zeroJacobian(Frame::kEndEffector, state);
Map<const Mat6x7> J(jacobian_array.data());
const Vec6 xdot = J * dq;

Map<const Mat4x4> T_EE(state.O_T_EE.data());
const Vec3 p_EE = T_EE.block<3, 1>(0, 3);
const Mat3 R_EE = T_EE.block<3, 3>(0, 0);
```

Listing A.4: Core impedance, coupled set-up, and torque composition.

```
const Vec3 e_p = desired.p_d - p_EE;
const Vec3 e_R = applyRotationalAxisMask(
```

```
    params, orientationError(R_EE, R_d_used), R_base_surface);

Vec6 dx;
dx.head<3>() = e_p;
dx.tail<3>() = e_R;
Vec6 dv;
dv.head<3>() = desired.pdot_d - pdot;
dv.tail<3>() = -omega;

const Vec6 wrench =
    computeCartesianImpedanceWrench(
        params, phase, Kp_used, Dp_used,
        KR_used, DR_used, R_base_surface,
        dx, dv, R_EE);

const Vec7 tau_nullspace = computeNullspaceTorque(
    params, model, state, J, dq);

AutomaticDisturbance disturbance;
if (phase == ControlPhase::kPoseHold &&
    params.disturbance_auto_enabled) {
  disturbance = computeAutomaticDisturbance(
      params, model, state, disturbance_force_direction_base,
      time - phase_start_time);
}

Array7 coriolis_array = model.coriolis(state);
Map<const Vec7> coriolis(coriolis_array.data());
const Vec7 tau_task = J.transpose() * wrench;
const Vec7 tau_cmd =
    tau_task + tau_nullspace + disturbance.tau + coriolis;
```

The null-space function is called in every non-manual phase. Manual guidance uses its separate Coriolis-plus-joint-damping branch. The disturbance term is non-zero only in the Cartesian pose-hold study of Section 4.6; it is zero in every surface-contact run.

Appendix B

Recorded Experimental Data

ORANGE — sufficient. The appendix records the signals actually used in the evaluation and identifies the common calibrated plane used by the controller and analysis.

The controller stores experimental signals in a preallocated ring buffer. This avoids file access and dynamic buffer growth inside the 1 kHz control callback. After the controller stops, the retained rows are written in chronological order to a CSV file.

B.1 Control Phases

ORANGE — sufficient. The numeric phase values are mapped directly to their recorded state names.

The integer `phase` identifies the active part of the experiment:

`phase`	0	1	2	3	4	5
state	`approach_orient`	`approach_descend`	`set_up`	`grind`	`hold`	`manual_guide`.

B.2 Signals Used in the Evaluation

ORANGE — sufficient. The table is a useful reproducibility record and keeps units, frames, and model-estimated quantities explicit.

Table B.1 lists the groups used to calculate the reported quantities. A suffix `_x,_y,_z` denotes base-frame components, and `_1.._7` denotes joint components.

Table B.1: Recorded signal groups used in the evaluation.

Column group	Meaning
`time`, `phase`	Run time and active control state.
`p_EE`, `p_d`	Measured and desired TCP positions [m].
`tool_contact`, `edge_target`	Selected contact point and its reference [m].
`tool_contact_offset_ee`	Offset of the selected contact point from the TCP, in EE coordinates [m].
`first_contact_tcp`, `first_contact`	TCP position and projected surface reference captured at the geometric clearance transition; the transition is not force triggered.
`e_p`, `e_R`, `pdot`, `pdot_d`, `omega`	Cartesian errors and velocities. During set-up, `e_R` is measured relative to the clearance orientation selected at the transition and held constant during set-up.
`angular_deviation_deg`	The pose-based alignment error retained for the consistency check of Appendix D.7, $\theta_{\text{align}} = \cos^{-1}(-n_{\text{Tool},0}^{\top} n_s)$, between the pose-based tool axis $n_{\text{Tool},0}$ and the flat target $-n_s$, with n_s the calibrated physical-plane normal stored in the controller configuration [°]. It is zero for a tool face parallel to the surface.
`angular_deviation_t1_deg`, `angular_deviation_t2_deg`, `angular_deviation_normal_deg`	The same angular deviation resolved on t_1, t_2, and n_s [°]. These recorded controller fields are not main evaluation quantities.
`p_CoC`, `r_eff`	CoC $p_{\text{TCP}} + r_c$ and the offset from it to the selected tool point, $p_{\text{Tool}} - p_c$, both in the base frame [m]. A zero lever leaves the centre on the TCP, so the pair is written for every run and no case has to be separated.

Table B.1 continued

Column group	Meaning
`t_align`	Time from the start of set-up at which the deviation first fell inside the configured tolerance and stayed there [s]; negative until that happens, and negative throughout for a run that never does. Observed only: reaching it does not end the phase.
`f`, `m`	Commanded Cartesian force [N] and moment [N m].
`external_force`, `external_moment`	Model-estimated external wrench from libfranka, expressed in the base orientation and referenced to stiffness frame K.
`contact_force_bias`, `contact_moment_bias`	External-wrench values captured at the clearance transition.
`force_after_contact`, `moment_after_contact`	The model-estimated wrench with those reference values already subtracted.
`push`	Set-up penetration reference or grind preload retained after set-up [m].
`nullspace_mode`, `sigma_current`	Selected null-space law and current smallest singular value $\sigma_{\min}$.
`sigma_plus`, `sigma_minus`, `sigma_difference`, `sigma_direction`	Smallest singular value $\sigma_{\min}$ at the two sampled configurations $q \pm \alpha_{\text{probe}} v_7$, their difference, and the selected sign. Recorded in every mode, including when no torque is commanded.
`nullspace_dq_1..7`	Projected joint velocity $\dot{q}_{\text{null}}$ [rad/s].
`tau_sigma_1..7`, `tau_sigma_norm`, `tau_nullspace_norm`	Singular-value torque τ_σ and total null-space torque τ_{null} [N m].
`nullspace_damping`	Active projected damping gain [N m s/rad].

Table B.1 continued

Column group	Meaning
`disturbance_scale`, `disturbance_force_base_x..z`, `disturbance_point_base_x..z`	Commanded hold-disturbance waveform, point force [N], and the base-frame position of the link point it acts at [m].
`tau_disturbance_1..7`	Equivalent joint torque $J_p^\top f_d$ in the internally commanded hold study [N m].
`tau_cmd_1..7`	Final commanded joint torque [N m].

B.3 Evaluation Quantities

ORANGE — sufficient. The short summary routes each reported metric to its recorded source without reintroducing unused fields.

The pose-based alignment error θ_{align} was calculated from the measured end-effector orientation, the calibrated tool normal $n_{\text{Tool,EE}}$, and the calibrated surface normal n_s. Its before value was taken at the start of set-up, and its after value from the settled set-up interval. It is retained only for the appendix consistency check. No other plane normal was substituted in the analysis. The main signed set-up rotation was calculated from `e_R` at the end of set-up and resolved on the surface tangents, as Equation (4.1) and Equation (4.2) define it.

Three markers of that same angle are written once into the set-up report rather than into every row. All three read θ_{align} and the clock alone. None feeds the phase exit or the command. `t_align_fraction` is the time at which θ_{align} first fell below a configured fraction of its value at first contact. Being a relative threshold, it is reached by conditions that end far from flat, which the absolute tolerance is not. `deviation_min` is the smallest θ_{align} the run reached, together with the time at which it was reached. Both are defined for every run, so a condition that never aligned still records how close it came. `align_status` reads `aligned` where θ_{align} settled inside the tolerance for the required hold time, and `not_aligned` otherwise. Runs that never settled are identified rather than discarded.

None of these determines whether the tool itself ended flat. That is calculated afterwards from the height of the TCP above the calibrated plane, $n_s^\top(p_{\mathrm{EE}} - p_s)$, against the height a flatly seated tool face would give. Only p_{EE} and the stored plane enter it. Unlike θ_{align}, it therefore carries no assumption about the rotation of the tool within the gripper.

For the internally commanded hold study, redundant motion is calculated by integrating the norm of the recorded projected joint velocity $\dot{q}_{\mathrm{null}}$ over the disturbance interval. The conditioning response is the final change in $\sigma_{\min}$, and Cartesian task retention is represented by the peak position-error norm $\|e_p\|$.

Appendix C

Controller Parameters

ORANGE — sufficient. The active geometry, phase gains, damping policy, timings, null-space values, and collision settings are collected in one reproducibility record.

This appendix lists the effective controller parameters used in the reported experiments. It is a reproducibility record for the implementation described in Chapters 3 and 4.

C.1 Surface and Tool Geometry

ORANGE — sufficient. The configured geometry and initial joint vector are stated with their exact keys and units.

The plane satisfies

$$n_s^\top (p - p_s) = 0, \qquad n_s = R_y(\beta_s) R_x(\alpha_s) e_z,$$

and the surface-frame column order is $[t_1, t_2, n_s]$. Table C.1 collects the corresponding surface and tool parameters.

Table C.1: Surface and tool parameters.

Quantity	Configuration key	Value
Surface point	`surface_point`	$[0.515267, -0.107172, 0.003108]\,\mathrm{m}$
Surface tilts	α_s, β_s	-1.585°, $+0.988^\circ$
First-tangent hint	`surface_tangent1_hint_base`	$[1, 0, 0]$
Tool axis in EE	`tool_axis_ee`	$[0.026387057, -0.006678514, 0.999629492]$
Signed alignment target	`tool_axis_target_sign`	$-n_s$
Spin about the tool axis, from t_1	`tool_target_offset_normal_deg`	90.0°
Tool-face centre in EE	`tool_contact_face_center_ee`	$[0, 0, 0.020]\,\mathrm{m}$
Tool face size	width $\times$ length	$0.040 \times 0.120\,\mathrm{m}$
Feature tie tolerance	`tool_contact_feature_tie_tolerance`	$10^{-4}\,\mathrm{m}$

C.1.1 Calibration Relations

For the seated surface pose, the nominal end-effector axis and the tool-face centre offset supplied the stored plane quantities,

$$n_s = R_{\mathrm{EE}}[0, 0, 1]^\top, \qquad p_s = p_{\mathrm{EE}} + R_{\mathrm{EE}} r_{\mathrm{face,EE}}. \tag{C.1}$$

The first tangent was the normalised projection of the base-frame x direction onto the plane, and $t_2 = n_s \times t_1$ completed the surface frame. For the tool-normal calibration, the fitted direction was

$$n_{\mathrm{Tool,EE}} = \underset{\|n\|=1}{\arg\min} \sum_{i<j} \left\|(R_{\mathrm{cal},i} - R_{\mathrm{cal},j})n\right\|^2 . \tag{C.2}$$

The first three seated yaw captures entered the fit. A fourth capture was held out as a consistency check. The sign of the fitted eigenvector was selected towards nominal $+Z_{\mathrm{EE}}$.

C.1.2 Initial Joint Configuration

The initial joint configuration used in the reported experiments is

$$
\begin{aligned}
q_{\text{init}} = [&0.014777,\ 0.210911,\ -0.009320,\ -2.284835, \\
&-0.002725,\ 2.493410,\ 0.780694]^\top \text{ rad.}
\end{aligned}
$$

C.2 Phase Sequence and Impedance

ORANGE — sufficient. The phase-specific gain frames and transition parameters reconcile the values stated across Chapters 3 and 4.

Table C.2: Phase and mode gains.

Phase / quantity	Active stiffness and frame	Configured damping
Approach translation	$[150, 150, 150]$ N/m, surface $[t_1, t_2, n_s]$	$[20, 20, 20]$ N s/m
Approach rotation	$[180, 180, 90]$ N m/rad, surface $[t_1, t_2, n_s]$	$[15, 15, 12]$ N m s/rad
Pre-set-up gate translation	$[2000, 2000, 350]$ N/m, surface $[t_1, t_2, n_s]$	$[10, 10, 175]$ N s/m
Pre-set-up gate rotation	$[5, 5, 50]$ N m/rad, surface $[t_1, t_2, n_s]$	$[10.01, 10.01, 10]$ N m s/rad
Set-up translation (compliance-centre source)	$[2000, 2000, 350]$ N/m in all surface-contact experiments, surface $[t_1, t_2, n_s]$	$[10, 10, 175]$ N s/m
Set-up rotation (compliance-centre source)	$[5, 5, 50]$ N m/rad, surface $[t_1, t_2, n_s]$	$[10.01, 10.01, 10]$ N m s/rad
Grind translation (decoupled)	as the set-up entry above, surface $[t_1, t_2, n_s]$	$[10, 10, 175]$ N s/m
Grind rotation (decoupled)	$[5, 5, 50]$ N m/rad, surface $[t_1, t_2, n_s]$	$[10.01, 10.01, 10]$ N m s/rad
Hold translation	2000 N/m, isotropic base	50 N s/m
Hold rotation	50 N m/rad, isotropic base	8 N m s/rad
Manual guidance	no Cartesian stiffness; joint coordinates	0.5 N m s/rad joint damping

The pre-set-up gate was disabled in all reported contact runs, so its defined gains were inactive. The enabled pre-grind gate remains in set-up and keeps the coupled pressed law rather than using the pre-set-up gate gains.

The reported experiments used the surface-frame translational gains listed above. The active timing and trajectory values are listed in Table C.3.

Table C.3: Phase-transition and trajectory parameters.

Quantity	Configuration key	Value
Orient minimum time	`approach_orient_min_time`	1.0 s
Tool-axis switch tolerance	`approach_orient_error_threshold`	0.016 rad
Spin switch tolerance	`approach_orient_spin_error_threshold`	0.009 rad
Orient timeout / maximum rate	`approach_orient_timeout`, `approach_orient_max_rate_deg`	8.0 s, 20°/s
Descent speed / maximum distance	`descend_speed`, `descend_max_distance`	0.1 m/s, 0.640 m
Surface clearance	`descend_surface_clearance`	0.020 m
Set-up minimum time / timeout	`setup_min_time`, `setup_timeout`	2.0 s, 5.0 s
Set-up moment-change threshold	`setup_moment_threshold`	150 N m
Set-up push speed / endpoint	`setup_push_speed`, `setup_push_end`	0.080 m/s, 0.240 m in all surface-contact experiments
Configured grinding sweep	`grind_sweep_enabled`	enabled along t_2, 0.030 m, 0.2 Hz; not entered in the reported runs
Pre-set-up / pre-grind gates	`pause_before_set_up`, `pause_before_grind`	disabled, enabled

C.3 Inertia-Scaled Damping and Centre of Compliance

Inertia-scaled damping follows Equations (3.15) and (3.19). It is enabled for approach, set-up, the configured grinding phase, and the pause gate. The approach factor is $\zeta = 1.9$, while set-up and grinding use $\zeta = 1.0$. The pause gate fits separate translational and rotational factors to its two configured damping targets. The configured maximum coefficient `auto_damping_max` is 400. The configured values in Table C.2 provide fallback damping for the phases using inertia-scaled damping. With `auto_damping_min_from_manual` left at zero, those manual values act only as that fallback and impose no lower bound on a calculated coefficient.

Standard Cartesian pose hold instead used its configured matrices directly. All twelve reported null-space runs recorded `hold_auto_damping=0`, with $D_p = 50 I_3\,\mathrm{N\,s/m}$ and $D_R = 8 I_3\,\mathrm{N\,m\,s/rad}$. The hold-damping values in Table C.2 are therefore active values rather than fallbacks.

For set-up, the source damping blocks are calculated from TCP task inertia Λ and then shifted to the selected CoC p_c. This preserves symmetry and positive semidefiniteness. The factor $\zeta = 1$ is the directional coefficient used by the online damping calculation.

The coupled 6×6 law is enabled only during set-up and only for conditions with a displaced CoC. The surface-fixed conditions of the definition-frame check in Appendix D.3 used a displacement defined in the calibrated surface frame. The other shifted conditions used tool-fixed displacements configured in end-effector coordinates. For the surface-fixed definition,

$$r_c = R_{\text{surface}}\, r_c^S, \qquad r_c^S = [\, r_{c,t_1},\ r_{c,t_2},\ r_{c,n}\,]^\top.$$

The tested definitions and components are listed in Table 4.4 and Appendix D. A tool-fixed lever remains constant in end-effector coordinates and rotates in the base frame with the end effector; a surface-fixed lever remains constant in surface coordinates. The controller transforms the selected definition to base coordinates during each set-up cycle.

C.4 Null-Space and Operating Parameters

ORANGE — sufficient. The selected mode, gains, probe settings, logging capacity, gripper state, and robot-side thresholds are stated explicitly.

The null-space settings are explained in Section 4.3.2. The final two rows of Table C.4 are Franka collision/reflex thresholds configured through the libfranka collision-behaviour interface. Robot-side monitoring applied these thresholds throughout the reported runs.

Table C.4: Null-space and operating parameters.

Parameter	Symbol / key	Value
Null-space mode	`nullspace_mode`	3 (projected damping and singular-value conditioning)
Null-space damping	d_{null}	2.0 N m s/rad
Sigma torque magnitude	k_σ	1.0 N m
Probe distance	α_{probe}	0.08 rad
Sigma difference deadband	δ_σ	10^{-6}
Relative SVD cutoff	ρ	10^{-4}
Recorded-data sampling / capacity	`log_every_n_cycles`, rows	every callback (1), 120000
Run duration / stop	`experiment_duration`	0 s: indefinite until `e+Enter`
Gripper grasp	width, speed, force	0.066 m, 0.050 m/s, 60 N
Tool-mount rotational play	observed pad-mount clearance	approximately $\pm 2^\circ$ about y_{EE}
Manual-guidance damping	`manual_guidance_damping`	0.5 N m s/rad
Joint-side collision threshold	acceleration/nominal	140 N m on each axis
Cartesian collision threshold	acceleration/nominal	240 N or N m on each channel

Appendix D

Supporting Contact Checks and Results

This appendix reports the complete numerical values behind the main contact figures and four supporting checks. The supporting checks comprise 54 runs across 18 independently counted settings, with three repetitions per setting. The calibrated geometry, phase sequence, Cartesian gains, null-space settings, and evaluation quantities were those of Sections 4.2 and 4.3 unless a changed condition is stated below.

D.1 Numerical Values for Main Cases A–D

The main chapter retains the comparison figures and interpretation. The tables below provide the corresponding numerical values, each as the arithmetic mean and sample standard deviation over the three repetitions of that setting.

Table D.1: Baseline contact response with the compliance centre at the TCP, mean ± SD over three repetitions. Nothing is varied: every setting holds $r_c = 0$ and the common Cartesian gains, so the measured rotation is the natural contact response against which Cases B to D are read.

Commanded Initial Orientation	CoC Position	Measured Set-Up Rotation $\Delta\theta_i$ [°]
No orientation offset	TCP	-0.97 ± 0.02
$+10°$ about t_1	TCP	$+7.57 \pm 0.01$
$-10°$ about t_1	TCP	-10.83 ± 0.06
$+10°$ about t_2	TCP	-1.63 ± 0.17
$-10°$ about t_2	TCP	-1.25 ± 0.19

Table D.2: Effect of rotational stiffness on the set-up rotation, mean ± SD over three repetitions. Varied: K_{R,t_i} about the commanded tangent. Held fixed: the compliance centre at the TCP, the commanded offset, and all other Cartesian gains. The 5 N m/rad row is the Case-A reference.

Commanded Initial Orientation	CoC Position	Varied Rotational Stiffness K_{R,t_i} [N m/rad]	Reference	Measured Set-Up Rotation $\Delta\theta_i$ [°]
$+10°$ about t_1	TCP	5	Case-A reference	$+7.57 \pm 0.01$
		15	—	$+7.29 \pm 0.01$
		50	—	$+2.73 \pm 0.03$
$+10°$ about t_2	TCP	5	Case-A reference	-1.63 ± 0.17
		15	—	-1.52 ± 0.31
		50	—	-0.48 ± 0.03

Table D.3: Effect of cross-axis translational stiffness on the set-up rotation, mean ± SD over three repetitions. Varied: the translational entry perpendicular to the commanded rotation axis. Held fixed: the compliance centre at the TCP, $K_{R,t_i} = 5\,\mathrm{N\,m/rad}$, and the commanded offset. The 2000 N/m row is the Case-A reference.

Commanded Initial Orientation	CoC Position	Varied Entry	K_p [N/m]	Reference	Measured Set-Up Rotation $\Delta\theta_i$ [°]
+10° about t_1	TCP	K_{p,t_2}	300	—	$+7.59 \pm 0.00$
		K_{p,t_2}	800	—	$+7.56 \pm 0.02$
		K_{p,t_2}	2000	Case-A reference	$+7.57 \pm 0.01$
+10° about t_2	TCP	K_{p,t_1}	300	—	-2.24 ± 0.14
		K_{p,t_1}	800	—	-1.78 ± 0.03
		K_{p,t_1}	2000	Case-A reference	-1.63 ± 0.17

Table D.4: Effect of tangential compliance-centre position on the set-up rotation, mean ± SD over three repetitions. Varied: the tangential CoC position. Held fixed: the Cartesian gains and the 10° commanded offset magnitude. The TCP column is the zero-position reference. The reported coordinate is r_{c,t_2} for commands about t_1 and r_{c,t_1} for commands about t_2. Consequently the displacement selected for a positive offset appears at +40 mm in the t_1 comparison and at −40 mm in the t_2 comparison.

Commanded rotation about t_1; coordinate r_{c,t_2}							
Offset	−40	−20	−10	**TCP**	+10	+20	+40 mm
+10°	0.18 ± 0.07	0.99 ± 0.44	7.45 ± 0.02	7.57 ± 0.01	7.67 ± 0.01	7.73 ± 0.01	7.87 ± 0.01
−10°	-11.30 ± 0.11	-11.18 ± 0.09	-11.14 ± 0.07	-10.83 ± 0.06	-10.36 ± 0.21	-5.01 ± 0.07	-0.09 ± 0.01

Commanded rotation about t_2; coordinate r_{c,t_1}							
Offset	−40	−20	−10	**TCP**	+10	+20	+40 mm
+10°	4.43 ± 0.07	-1.12 ± 0.09	-1.31 ± 0.13	-1.63 ± 0.17	-1.85 ± 0.36	-1.95 ± 0.47	-4.51 ± 2.36
−10°	4.68 ± 0.05	-1.28 ± 0.05	-1.29 ± 0.01	-1.25 ± 0.19	-1.29 ± 0.14	-1.51 ± 0.25	-10.51 ± 0.71

D.2 Orientation-Offset Magnitude

The 5° and 10° commanded offsets were compared with the CoC at the TCP and at the selected 40 mm tangential position. These four independently counted settings gave 12 runs.

Table D.5: Effect of the commanded orientation-offset magnitude. Varied: the offset magnitude. Held fixed: the Cartesian gains. Reference: the TCP-centred condition of each pair. The last column gives the change the selected displacement produced.

Rotation Direction	Varied Commanded Offset	TCP-Centred $\Delta\theta_i$ [°]	Selected 40 mm CoC $\Delta\theta_i$ [°]	Effect of the CoC [°]
About t_1	5°	+2.55	+2.70	+0.15
About t_1	10°	+7.57	+7.87	+0.30
About t_2	5°	−1.40	+5.91	+7.31
About t_2	10°	−1.63	+4.43	+6.06

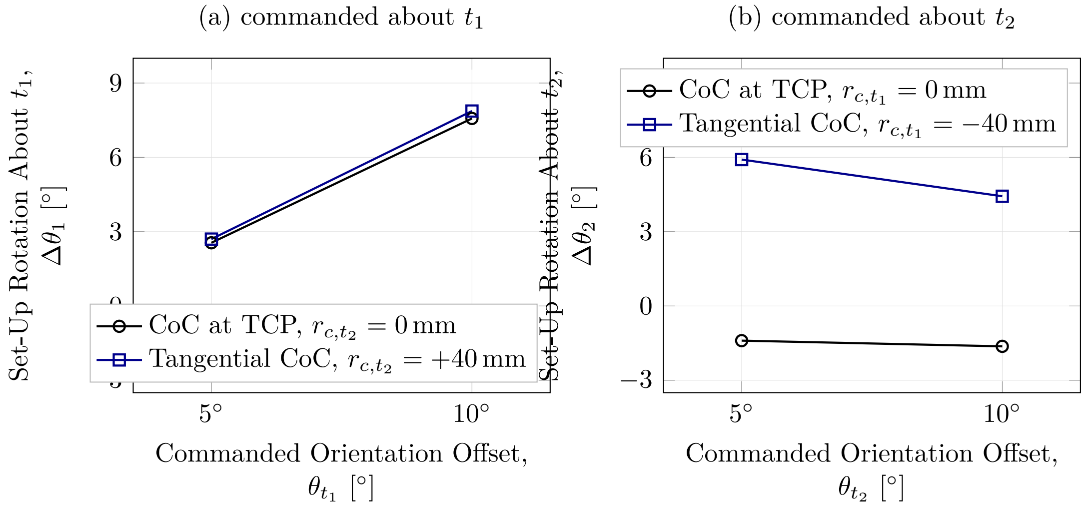

Figure D.1: Set-up rotation by commanded orientation-offset magnitude.

The response did not scale proportionally with the commanded offset. About t_2, the selected lever produced +5.91° at 5° and +4.43° at 10°, as shown in Figure D.1. The response therefore depends on the complete contact condition rather than on the offset magnitude alone.

D.3 Compliance-Centre Definition Frame

This check compared tool-fixed and surface-fixed definitions of the same nominal 40 mm tangential displacement. The tool-fixed vector rotated with the end effector. The surface-fixed vector remained constant in the calibrated frame $[t_1, t_2, n_s]$. Both vectors were transformed to the base frame during each control cycle before the stiffness and damping matrices were shifted to the TCP.

The supporting-check settings were the tool-fixed displacement without an orientation offset and the surface-fixed displacement with no offset or with a $+10°$ offset about t_1. These three settings gave 9 runs. The shared tool-fixed $+10°$ condition is counted with the $+40$ mm reference of Case D.

Table D.6: Effect of the frame in which the displacement is held constant. Varied: the definition frame, at the same nominal 40 mm tangential magnitude. Held fixed: the Cartesian gains and the commanded offset. Reference: the tool-fixed column, which is the Case-D $+40$ mm condition.

Commanded Initial Orientation	Tool-Fixed Definition $\Delta\theta_1$ [°]	Surface-Fixed Definition $\Delta\theta_1$ [°]
None	-0.06 ± 0.01	-1.05 ± 0.02
$+10°$ about t_1	$+7.87 \pm 0.01$	$+0.13 \pm 0.01$

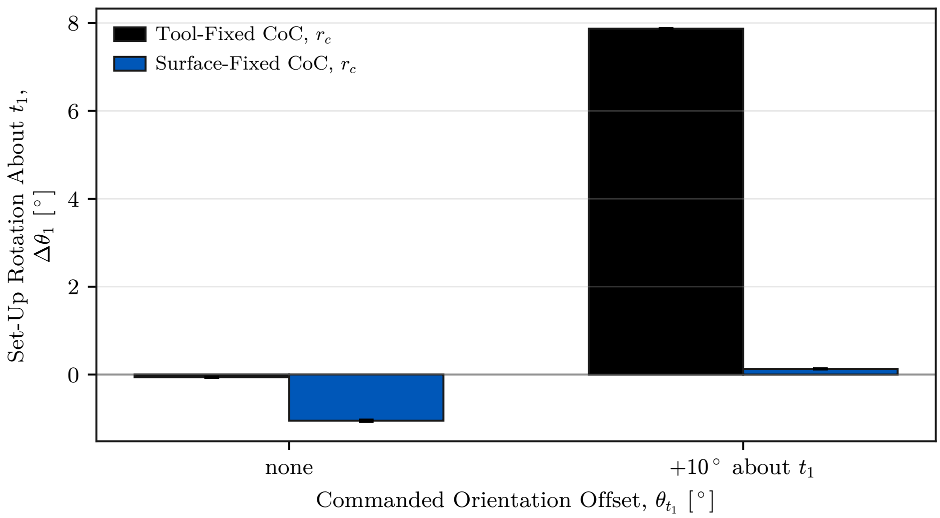

Figure D.2: Set-up rotation for tool- and surface-fixed displacement definitions.

The two definitions produced markedly different responses. Table D.6 pairs them at each command, and Figure D.2 shows the separation. For the $+10°$ command about t_1, the tool-fixed definition gave $+7.87 \pm 0.01°$ of set-up rotation. The surface-fixed definition gave $+0.13 \pm 0.01°$ and was classified as tilted by TCP height. At zero commanded offset, the two definitions produced $-0.06 \pm 0.01°$ and $-1.05 \pm 0.02°$.

A non-zero CoC displacement is therefore not fully characterised by its magnitude alone. The displacement direction and the frame holding it determine how the vector moves relative to the contact geometry. Returning the centre to the TCP removes that direction and definition-frame dependence together with the point-shift coupling.

D.4 Tool-Axis Displacement

This check separated the distance between the CoC and TCP from the direction of the displacement. The centre was moved by −40, −20, −10, +10, +20, and +40 mm along the tool axis at a fixed +10° command about t_1. These six settings gave 18 runs. The shared zero-displacement condition is counted with the Case-A reference.

Table D.7: Effect of displacing the compliance centre along the tool axis. Varied: the tool-axis position. Held fixed: the Cartesian gains and the +10° command about t_1. The 0 mm column is the Case-A reference.

Measured Response	−40	−20	−10	**0**	+10	+20	+40 mm
$\Delta\theta_1$	+7.26	+7.49	+7.58	+7.57	+7.58	+7.55	+7.60
Standard deviation	0.13	0.01	0.01	0.01	0.01	0.03	0.04

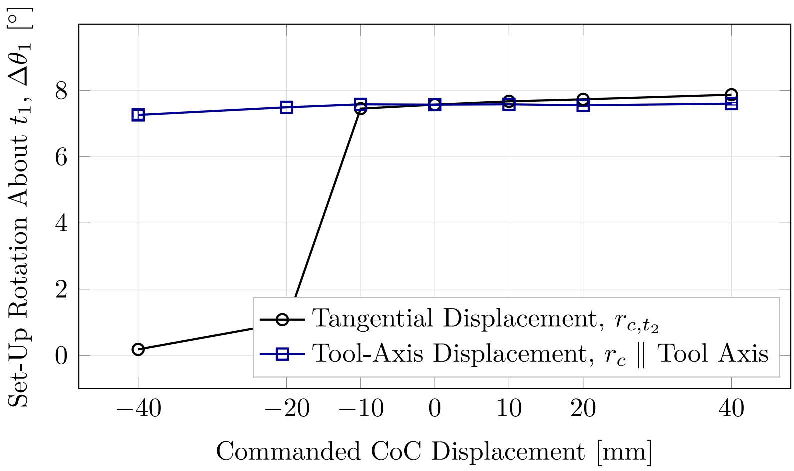

Figure D.3: Tangential and tool-axis centre displacement at the same command.

The tool-axis response varied weakly across the tested range. Table D.7 lists the seven positions, including the shared zero-displacement reference. The signed set-up rotation changed from +7.26 to +7.60° between −40 and +40 mm, a span of 0.34°. The purely tangential displacement spanned 7.73° over the same interval in Case D, as shown in Figure D.3. Every tool-axis position was classified as flat, and the response was not strictly monotonic.

As Equation (2.47) shows, only the tangential component of the CoC displacement contributes to the moment generated by the commanded normal force. Here the displacement was defined along the tool axis rather than along a surface tangent.

Let α_{axis} denote the acute inclination of the tool axis relative to the inward surface-normal direction $-n_s$, which is the direction the tool axis takes when the face is flat on the surface. The check used a $+10°$ commanded orientation offset about t_1, so the tool axis was nominally inclined by $10°$ from $-n_s$. A displacement along that inclined axis therefore contained both a normal and a tangential component. Its tangential magnitude was

$$\|r_{c,t}\| = \|r_c\| \sin \alpha_{\text{axis}}. \tag{D.1}$$

At the largest tested tool-axis displacement, $\|r_c\| = 40\,\text{mm}$. Evaluating the projection at the nominal inclination, to two decimal places, gives

$$\|r_{c,t}\| = (40\,\text{mm}) \sin(10°) = 6.95\,\text{mm}.$$

The 40 mm value is the total configured CoC displacement along the tool axis. The 6.95 mm value is only its projection onto the surface tangent plane. This effective tangential lever is considerably smaller than the 40 mm purely tangential displacement of Case D. The difference is consistent with the weak variation in Figure D.3. Tangential force components, finite rotation, changing contact geometry, and tool-mount compliance were not isolated by this check.

D.5 Intermediate Tangent-Plane Directions

This check examined whether the direction rule of Case D also held between the principal surface tangents. For a commanded rotation with surface components $(\theta_{t_1}, \theta_{t_2})$, the selected tangential displacement was

$$r_{c,t} = \frac{\|r_{c,t}\|}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}} \left(\theta_{t_1} t_2 - \theta_{t_2} t_1\right), \tag{D.2}$$

where $\|r_{c,t}\| = 40\,\text{mm}$. The selected vector was perpendicular to the commanded rotation in the tangent plane. Its sign made the press-induced moment act in the required alignment direction. The fixed comparison retained the vector selected for the t_1 command.

Table D.8: Commanded directions and the tangential compliance-centre vectors tested against them. Every setting was configured tool-fixed in end-effector coordinates; the vectors given are the surface-frame components they realise in the flat target orientation.

Commanded Rotation Direction	Displacement Setting	Commanded Rotation Components, $[\theta_{t_1}, \theta_{t_2}]$ [°]	CoC Components, $[r_{c,t_1}, r_{c,t_2}, r_{c,n}]$ [mm]
About t_1	selected	$[+10.0, 0.0]$	$[0, +40.0, 0]$
$-45°$ from t_1	selected	$[+7.1, -7.1]$	$[+28.3, +28.3, 0]$
$+45°$ from t_1	selected	$[+7.1, +7.1]$	$[-28.3, +28.3, 0]$
About t_2	selected	$[0.0, +10.0]$	$[-40.0, 0, 0]$
$-45°$ from t_1	fixed at the t_1 selection	$[+7.1, -7.1]$	$[0, +40.0, 0]$
$+45°$ from t_1	fixed at the t_1 selection	$[+7.1, +7.1]$	$[0, +40.0, 0]$
About t_2	fixed at the t_1 selection	$[0.0, +10.0]$	$[0, +40.0, 0]$

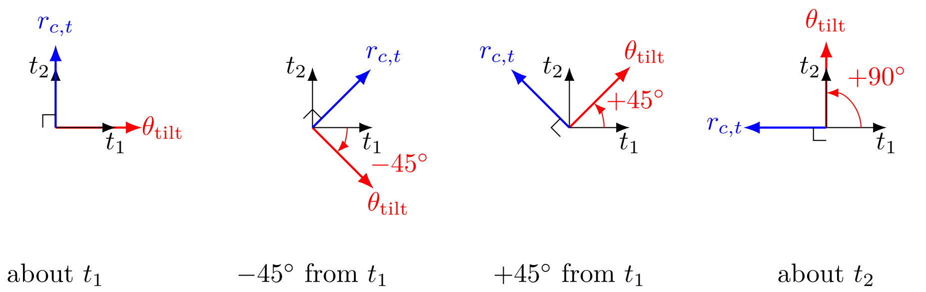

Figure D.4: Selected tangential displacement for each commanded rotation direction.

The principal-axis selected conditions are counted with Case D. The five supporting-check settings were both displacement definitions at each diagonal direction and the fixed displacement for the t_2 command. Each was repeated three times, giving 15 runs. The vectors in Table D.8 are expressed in the surface frame at the flat target orientation. The corresponding non-zero controller settings were held tool-fixed in end-effector coordinates.

Table D.9: Direction-rule comparison at the common magnitude $\|r_{c,t}\| = 40$ mm. Varied: the commanded rotation direction. Held fixed: the Cartesian gains and the displacement magnitude. Reference: the fixed t_1 selection, retained across all four directions.

Commanded Rotation Direction [°]	Selected Displacement $[r_{c,t_1}, r_{c,t_2}]$ [mm]	Selected $\Delta\theta_i$ [°]	Fixed at t_1 $\Delta\theta_i$ [°]
About t_1	$[0, +40.0]$	$+7.87$	$+7.87$
$-45°$ from t_1	$[+28.3, +28.3]$	$+4.46$	$+4.39$
$+45°$ from t_1	$[-28.3, +28.3]$	$+4.85$	$+4.80$
About t_2	$[-40.0, 0]$	$+4.43$	-1.61

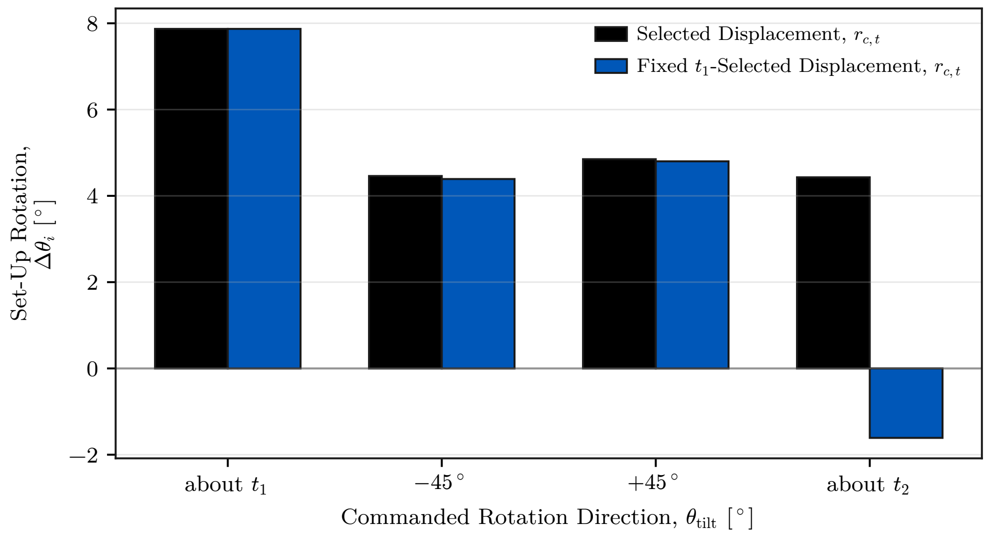

Figure D.5: Set-up rotation with the direction-selected and fixed displacements.

At the two diagonal directions, the selected and fixed displacements differed by only 0.07° and 0.05°, within the observed scatter. The fixed displacement retained a substantial projection along the selected direction in both conditions. The clearest separation occurred about t_2, where the selected displacement gave +4.43° and the fixed displacement gave −1.61°.

The measurements support the direction rule of Section 2.7.2 over the four tested tangent-plane directions. They do not establish a single non-zero displacement that suits all directions. Where the projection of the fixed displacement onto the selected direction vanished, the responses differed by 6.04°.

The remaining sections resolve summary statements of Chapter 5 into the settings and repetitions behind them, and compare the primary response metric with the supporting one.

D.6 Conditions Classified as Tilted by TCP Height

The TCP-height criterion is retained here as a supporting geometric check and does not enter the primary response comparisons of Chapter 5. All five Case-A settings, including the zero-orientation-offset condition, satisfied this criterion. This classification is not a direct measurement of the physical tool-face orientation. The mounting play was characterised in the unloaded condition, and the instantaneous relative tool–gripper rotation was not tracked separately during the contact runs.

Of the 55 tested parameter settings, 49 satisfied the TCP-height flatness criterion. Table D.10 lists the six that did not, with the excess TCP height above the flat-tool height.

Table D.10: Conditions classified as tilted by TCP height.

Condition	Excess height [mm]	$\Delta\theta_1$ [°]
Case B, 50 N m/rad about t_1	3.6	+2.73
Case D, −20 mm at +10° about t_1	5.3	+0.99
Case D, −40 mm at +10° about t_1	6.1	+0.18
Case D, +20 mm at −10° about t_1	5.2	−5.01
Case D, +40 mm at −10° about t_1	9.3	−0.09
Definition-frame check, surface-fixed at +10° about t_1	5.9	+0.13

The signed set-up rotation does not use the calibrated tool normal. It is obtained from the measured end-effector motion and resolved in the surface frame, which makes it the direct quantity for comparing controller-induced rotation between settings.

D.7 Pose-Based Alignment Consistency Check

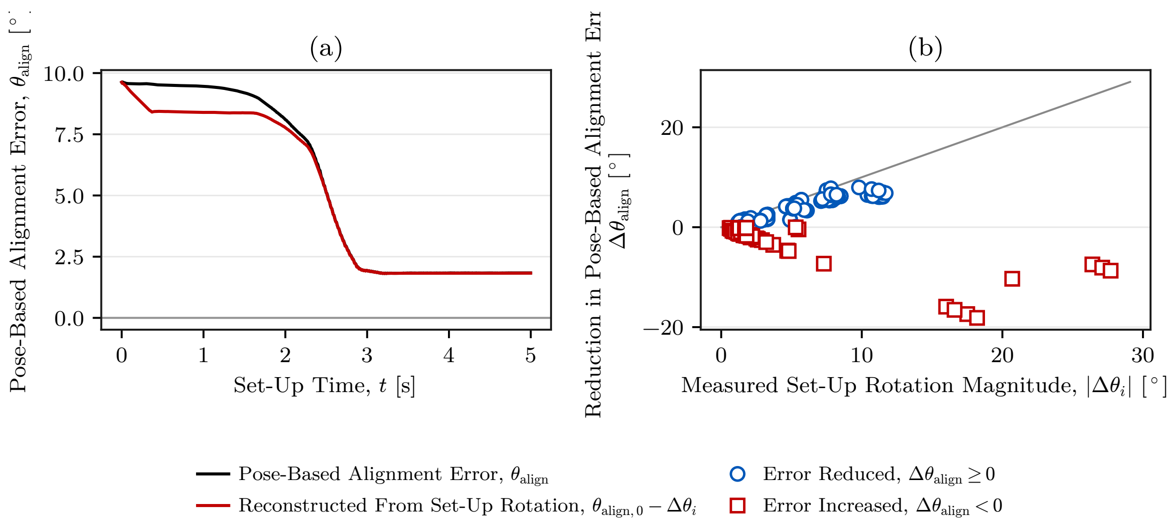

Figure D.6: Pose-based alignment estimate against measured set-up rotation.

The main results use the signed end-effector set-up rotation $\Delta\theta_i$. This appendix compares that direct motion measurement with a secondary alignment estimate reconstructed from the calibrated tool normal. The pose-based alignment error is

$$\theta_{\mathrm{align}}(t) = \cos^{-1}\Big(-n_{\mathrm{Tool},0}(t)^{\top} n_s\Big)\,, \qquad n_{\mathrm{Tool},0} = R_{\mathrm{EE}} n_{\mathrm{Tool,EE}}. \tag{D.3}$$

It is unsigned and depends on the assumed fixed relation between the tool and end effector. Its reduction over set-up is written

$$\Delta\theta_{\mathrm{align}} = \theta_{\mathrm{align,before}} - \theta_{\mathrm{align,after}}, \tag{D.4}$$

which is used in this appendix only and does not enter the notation of the main chapters. It and $\Delta\theta_i$ are defined differently and are not interchangeable.

Panel (a) plots the pose-based error for one representative trial and the value reconstructed from the measured rotation since the start of set-up. Panel (b) reduces the comparison to one point per evaluated trial. Each point carries the magnitude of the measured set-up rotation and the signed reduction of the pose-based error. Where the pose-based error decreased, its reduction followed the same scale as the measured rotation for most trials. Conditions in which the pose-based error increased appear below zero and show why the two quantities are not interchangeable.

The pose-based alignment quantity depends on the assumed tool-to-gripper relation, whereas the set-up rotation is obtained directly from the measured end-effector motion. Their agreement therefore provides an internal consistency check but does not constitute an independent measurement of the physical tool orientation. The approximately $\pm 2^\circ$ of mounting play was measured unloaded and was not characterised as an uncertainty bound under contact load.